

Is Liquidity Risk Priced in Partially Segmented Markets?*

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Abstract

We analyze the impact of liquidity costs and market segmentation on asset pricing. The freely traded securities command a global market risk premium and world liquidity risk premia whereas the securities that can be held by only a subset of investors command additionally a conditional local market risk premium and conditional local liquidity risk premia. Our model provides a formal framework for testing important issues such as liquidity levels and risks in a realistic world market setting and for examining the interaction between investability and illiquidity. Empirical test results for a sample of 21 emerging markets strongly support theoretical predictions.

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1 Introduction

Liquidity costs and their uncertain variation affect the willingness and the ability of foreign investors to invest in emerging markets (EMs). Unfortunately, most of the asset pricing models under barriers to portfolio flows do not take into account liquidity concerns¹. We analyze the effect of illiquidity cost and systematic liquidity risk factors on the pricing of EM securities by developing a formal international asset pricing model (IAPM) that takes into account cross-border investment barriers as well as random transaction costs. Our paper is close to the work of [Lee \(2011\)](#) and [Bekaert, Harvey, and Lundblad \(2007\)](#) [BHL hereafter].² Whereas Lee adopts an econometric specification to obtain his testable equations, BHL analytically derive stylized pricing equations under market integration and segmentation. Our derivations are similar to BHL under the two polar cases. However, the pricing equation we obtain under mild segmentation allows us to test important issues such as liquidity levels and risks in a realistic world market setting and for examining the interaction between investability and illiquidity. Although our empirical set up focusses on EMs, it is applicable to many global markets characterized by investable and non-investable segments.

Our new liquidity IAPM exploits the theoretical insights of international frameworks under partial segmentation. In a two country framework, we assume that all domestic securities are investable and hence can be freely traded by all investors whereas the foreign market consists of two segments: (a) a non-investable set of securities that can be held only by local investors³, and (b) an investable set that can be held by all investors. We include random transaction costs on both investable and non-investable securities⁴. We show that the investable securities are priced as in [Acharya and Pedersen \(2005\)](#) [AP hereafter] but the market risk and liquidity shocks are aggregated at the world level. Also, a Jensen's inequality term that depends on the security's local liquidity obtains as in BHL because the model is formulated using log returns which are assumed to be normally distributed. Non-investable securities command additionally a conditional market risk premium and conditional liquidity risk premiums. The conditional market risk is a

¹See for example, [Stulz \(1981\)](#), [Errunza and Losq \(1985\)](#) [hereafter EL], [Eun and Janakiraman \(1986\)](#), [de Jong and Roon \(2005\)](#), [Chaieb and Errunza \(2007\)](#).

²BHL also suggest that liquidity effects should be more important than that observed in the United States.

³Non-investable assets cannot be or are not traded by domestic investors due to severe explicit and/or implicit barriers. Explicit barriers include legal restrictions on ownership, foreign exchange transactions, repatriation of profits, whereas implicit barriers encompass institutional, informational, governance, and market development variables.

⁴Liquidity could itself be endogenously determined by other implicit barriers such as asymmetric information (see [Johnson, 2006](#); [Huang and Wang, 2009](#), among others).

residual risk that stems from imperfect spanning of the returns on the non-investable segment of the foreign market by returns on substitute assets ⁵. All of the sensitivity of returns to aggregate local market risk, sensitivity of returns to aggregate local market liquidity, the sensitivity of liquidity to aggregate local market risk, and the local commonality in liquidity are conditional on returns on the substitute assets net of the illiquidity costs. The prices of global market risk and of global liquidity risk are the same. Similarly, the prices of local market risk and local liquidity risk are the same. Thus, our model extends the well-established theoretical asset pricing literature regarding the importance of liquidity for U.S. and other integrated developed markets (Amihud and Mendelson, 1986; Pastor and Stambaugh, 2003; Acharya and Pedersen, 2005, see, for example,). AP derive a liquidity-adjusted CAPM and obtain three liquidity related betas: commonality in liquidity as in Chordia, Roll, and Subrahmanyam (2000), return sensitivity to aggregate liquidity as in Pastor and Stambaugh (2003), and liquidity sensitivity to market return.

The second key contribution of the paper is a novel empirical set up that allows the estimation of the diversification portfolios, DPs in a conditional set up. DP is the dynamic replicating portfolio obtained by projecting the non-investables returns on substitute assets returns. The weights of the DP are time-varying and computed from the conditional covariance matrices. To estimate the conditional covariance matrices we proceed in two steps. We first fit an AR(1)-NGARCH(1,1) on the assets returns net of transaction costs and liquidity proxy to estimate the variances. In a second step, for each investable portfolio and non-investable portfolio, we estimate a bivariate dynamic normal copula (DNC) as in Christoffersen and Langlois (2013). We then use the conditional variances obtained from the AR(1)-NGARCH(1,1) models and the conditional correlations obtained from the DNC model to obtain the DP weights.

Our work adds to the vast empirical literature on the relation between the average stock returns and liquidity documented in several U.S. studies (see, for example, Brennan, Chordia, and Subrahmanyam, 1998), commonality in liquidity (see for example, Chordia et al., 2000; Hasbrouck and Seppi, 2001; Huberman and Halka, 2001; Koch, Ruenzi, and Starks, 2014, for evidence from the U.S.)⁶, as well as a number of empirical papers that investigate liquidity effects in international

⁵Substitute assets consist of all assets that can be freely traded by domestic investors including the world market portfolios, closed-end country funds, exchange traded funds, ADRs and GDRs.

⁶Additionally, there is ample evidence that liquidity risk is unconditionally priced for the U.S. market although the contribution of the liquidity risk premium varies across studies and depends on the proxy used for liquidity, see for example, Pastor and Stambaugh (2003), Acharya and Pedersen (2005), Sadka (2006), Korajczyk and Sadka (2008),

markets, (see, for example, Domowitz, Glen, and Madhavan, 2001; Lesmond, 2005; Bekaert et al., 2007; Lee, 2011; Goyenko and Sarkissian, 2014; Karolyi, Lee, and van Dijk, 2012; Amihud, Hameed, Kang, and Zhang, 2015)⁷. The above studies underscore the importance of liquidity in the U.S. and international markets. Our focus on pricing liquidity risk for investables and non-investables within the context of a formal asset pricing model that accounts for the partially segmented nature of EMs, along with the flexible empirical methodology used in the estimation, sets the paper apart from the earlier literature.

To test our asset pricing model, we need to define the investable and non-investable set of securities. We follow Karolyi and Wu (2012) and use cross-listing to characterize investable stocks.⁸ We also need a measure of illiquidity costs. In the theoretical set up, the exogenous illiquidity cost of a security is modeled as the per trade cost. The illiquidity cost could then be measured with the bid-ask spread. But the data are available only for short sample periods and for a small set of emerging countries limiting their use for validation of the asset pricing model and for studying the effect of illiquidity on expected returns. Alternatively, we could use the Amihud (2002) absolute return to volume ratio which is related to measures of price impact and fixed trading costs or the incidence of zero returns of Lesmond, Ogden, and Trzcinka (1999) which could proxy search costs.⁹ As we require a measure of the cost of trade to test our asset pricing model, we need to

Kim and Lee (2014). Watanabe and Watanabe (2008) investigate the dynamic nature of illiquidity premium and show that liquidity betas and liquidity risk premiums are larger when investors face a higher level of preference uncertainty. Hagstromer, Hansson, and Nilsson (2011) test a conditional version of the AP liquidity-adjusted CAPM and find significant time-variation in the illiquidity premium but commonality risk is the least important component.

⁷Specifically, Domowitz et al. (2001) document that transaction costs in emerging markets are significantly higher than in developed markets. Lesmond (2005) shows that liquidity costs are higher in countries with weak legal enforcement. BHL find that illiquidity predicts returns in emerging markets and that the level of liquidity is more important than systematic liquidity risk premiums. Based on an econometric model in an unconditional set up, Lee (2011) finds that liquidity risk is priced. Goyenko and Sarkissian (2014) analyze the impact of changes in the liquidity of US Treasury bills on global equity returns. Karolyi et al. (2012) show that commonality in liquidity is greater for emerging markets than for developed markets though it trended down for many countries. They also find that commonality in liquidity increases with capital inflows, while it decreases with market openness. Amihud et al. (2015) provide international evidence on the positive relation between expected return and illiquidity, using data from 45 countries including 19 emerging markets. See also Amihud and Mendelson (2015) for a review of the work on the effect of liquidity on expected returns.

⁸Although BHL do not formally incorporate the impact of changing market structure on pricing of liquidity risk, they do suggest that expected returns should decline post market liberalization which is related to investability. They also show that liquidity risk premium remain significant after market liberalization. Indeed, they define liberalization as the proportion of local market capitalization not subject to foreign ownership restrictions.

⁹Lesmond et al. (1999) show that the proportion of zero returns measure is highly correlated with the effective spread. This measure has been applied in a number of recent papers. See for example, BHL, Lesmond, Schill, and Zhou (2004), Lang, Lins, and Maffett (2012), and Mei, Scheinkman, and Xiong (2009) among others. Levine and Schmukler (2006) use both the proportion of zero returns and the Amihud's measure to examine the relationship between home market liquidity and cross listed trading across 31 home countries. Amihud (2002)'s widely applied price impact measure has been recently used in Karolyi et al. (2012) to analyze the commonality patterns of liquidity,

scale these two proxies as in, for example, AP. The paucity of effective bid-ask spread data for emerging markets limits such computation.¹⁰ Hence, we use the [Corwin and Schultz \(2012\)](#) high-low spread measure as proxy for the bid-ask spread. While it is a good proxy for the cost of a round-trip transaction of small size, this measure does not capture other dimensions of liquidity such as market depth, resilience and search costs.

We run the asset pricing tests in a conditional set up. We find significant and time-varying price of world covariance risk. Its magnitude of two is very plausible for the world aggregate risk aversion. The cross-sectional average of the price of local conditional covariance risk is about five with significant variation across markets. The price of local risk is proportional to the differential in risk aversion between an EM average investor and the world average investor and hence again seems economically plausible.

Our results suggest that the contribution of world liquidity risk premium to world risk premium and of local liquidity risk premium to local risk premium are on average small but the liquidity risk premiums contribution is time varying and higher in crises periods. The covariation of returns to world market illiquidity is the dominant factor both on average and in crisis periods characterized by significant reductions in liquidity.

Finally, we demonstrate the robustness of our results to various sensitivity exercises. Estimating the prices of risk from net returns while leaving the impact of expected liquidity costs unrestricted as in AP, and assuming all EM stocks are non-investable, do not change our base results in any meaningful way.

The article is organized as follows. Section 2 models the liquidity risk effect on expected returns in partially segmented markets. Section 3 outlines the empirical methodology. Section 4 presents data and summary statistics. Section 5 reports the results, and examines the statistical and economic significance of the liquidity risk premiums. Section 6 presents a robustness check using alternative specifications of the asset pricing equations and of the non-investable set. Section 7 concludes.

returns, and turnover across 40 countries. [Korajczyk and Sadka \(2008\)](#), [Mancini, Rinaldo, and Wrampelmeyer \(2013\)](#) and [Kim and Lee \(2014\)](#) use a latent liquidity measure constructed from a principal component across different liquidity measures. See also [Fong, Holden, and Trzcinka \(2014\)](#) for a recent review of the different liquidity proxies used in international studies.

¹⁰See BHL for a detailed discussion of trading costs data issues in EMs.

2 The asset pricing model

We postulate a two country global capital market: the domestic country D and the foreign country F . We label all securities that can be freely traded by all investors as investable and those that can be held by only foreign investors as non-investable. We can view the domestic market as a well-developed market (such as the United States) that is open to all investors and the foreign market as an EM where foreign participation is limited due to explicit and/or implicit barriers. In this setting, domestic (U.S.) investors can hold domestic securities and the investable segment of the foreign market whereas foreign (EM) investors can freely trade in the domestic and the foreign market. This characterization of the global market is fairly realistic. Of course, a market structure, in which both countries face some segmentation, is more attractive. However, our simplification makes understanding the forces at work easier without much loss of generality. ¹¹

2.1 Assumptions and notations

We use the subscript $i(n)$ as a generic index to represent the investable (non-investable) securities, bold letters to denote vectors and capital letters to denote matrices or portfolio level variables.

- **A1: Market Structure.** There are N risky assets partitioned as follow: The first N_I securities are the investable securities indexed by $i = 1, \dots, N_I$, and the second N_N assets are the non-investable securities indexed by $n = 1, \dots, N_N$. The vector of log-returns can be partitioned as $\mathbf{r}_t \equiv \begin{pmatrix} \mathbf{r}_{i,t} \\ \mathbf{r}_{n,t} \end{pmatrix}$. The vector of market values at time t , $\mathbf{M}_t$, is similarly partitioned as $\mathbf{M}_t \equiv \begin{pmatrix} \mathbf{M}_{i,t} \\ \mathbf{M}_{n,t} \end{pmatrix}$. $M_{N,t}$ is the market capitalization of non-investables at time t defined as $M_{N,t} = \sum_{n=1}^{N_N} M_{n,t}$. Agents can also borrow and lend at the domestic risk-free log-rate of return r_f , which is exogenous.
- **A2: Agents.** Every period, the economy has two agents with CRRA preferences. The first agent is domestic and is restricted to investing only in investable securities. The second agent is foreign and can invest in both non-investable and investable securities. Agents live for one

¹¹Indeed, as shown by [Chaieb and Errunza \(2007\)](#), the results carry through to the more general market structure with the addition of another premium at equilibrium.

period, invest at time t , and consume at $t+1$. Hence both investors have the same one-period trading horizon and holding period i.e. the trading frequency and holding period are set exogenously. This is a strong assumption. Allowing for endogenous trading frequency will result in less trading for illiquid assets. [Constantinides \(1986\)](#) shows that transaction costs might then have second-order effects on liquidity premia.

- **A3: Transaction costs.** Transaction costs denoted by the N -by-1 vector C_t are random and include bid-ask spread as well as search costs. They are modeled as the per-share cost of buying and selling an investable or non-investable security j . Short-selling is allowed. Both long and short holders pay transaction costs. As in AP, the one-period investors investing at $t-1$ pay transaction costs proportional to the current price when closing the position at time t . The net log-returns of the agents can then be written as $r_{j,t}^{net} = r_{j,t} - c_{j,t}$ for the investors who are long in asset j and as $r_{j,t}^{net} = -r_{j,t} - c_{j,t}$ for investors who are short in asset j , where $c_{j,t} = -\ln(1 - C_{j,t})$ (see [Bongaerts, Jong, and Driessen, 2011](#), for a similar assumption). We assume that the vector of net log-returns $\mathbf{r}_t^{net}$ is normally distributed and denote Σ_t as its N -by- N variance-covariance matrix which can be partitioned as $\Sigma_t \equiv \begin{pmatrix} \Sigma_{ii,t} & \Sigma_{in,t} \\ \Sigma_{in,t}^\top & \Sigma_{nn,t} \end{pmatrix}$, where $\Sigma_{ii,t}$ is the variance-covariance matrix of the investable assets, $\Sigma_{nn,t}$ is the variance-covariance matrix of the non-investable securities, and $\Sigma_{in,t}$ is the covariance matrix between investable and non-investable securities.
- **A4:** The returns are measured in domestic currency, the reference currency. We follow the asset pricing literature on barriers to cross-border investments and assume that the purchasing power parity holds (see, for example, [Stulz, 1981](#)).

2.2 Asset demands

The domestic investor d can invest in the riskless bond and the N_I investable risky assets. Let the vector $\boldsymbol{\omega}_{d,t}$ denote the fraction of his wealth invested in the risky assets at time t and γ^d his coefficient of relative risk aversion. Investor d maximizes at time t the utility of his terminal wealth W_{t+1}^d , that is,

$$\max_{\boldsymbol{\omega}_{d,t}} E_t \left[\left(\frac{W_{t+1}^d}{1 - \gamma^d} \right)^{1 - \gamma^d} \right]$$

subject to the budget constraint $W_{t+1}^d = \left(1 + R_{P,t+1}^d\right) W_t^d$ where $R_{P,t+1}^d = \boldsymbol{\omega}_{d,t}^\top \left(e^{\mathbf{r}_{i,t+1}^{net}} - e^{r_f} \boldsymbol{\iota}_{N_I}\right) + e^{r_f}$ is the simple net return on the portfolio of investor d and $\boldsymbol{\iota}_{N_I}$ is an $(N_I \times 1)$ vector of ones. Assuming the portfolio return is log-normal and using the [Campbell and Viceira \(2002\)](#) discrete-time approximation to relate asset log-returns to portfolio log-returns¹², the optimization can be restated as,

$$\max_{\boldsymbol{\omega}_{d,t}} \boldsymbol{\omega}_{d,t}^\top E_t \left[\mathbf{r}_{i,t+1} - \mathbf{c}_{i,t+1} - r_f \boldsymbol{\iota}_{N_I} + \frac{1}{2} \boldsymbol{\sigma}_{i,t}^2 \right] - \frac{1}{2} \gamma^d \boldsymbol{\omega}_{d,t}^\top \Sigma_{ii,t} \boldsymbol{\omega}_{d,t}$$

where $\boldsymbol{\sigma}_{i,t}^2$ is the vector containing the variance of investable asset net log-returns i.e. the diagonal elements of $\Sigma_{ii,t}$. From first order conditions, the solution for the vector of optimal portfolio weights is,

$$\boldsymbol{\omega}_{d,t} = \frac{1}{\gamma^d} \Sigma_{ii,t}^{-1} E_t \left[\mathbf{r}_{i,t+1} - \mathbf{c}_{i,t+1} - r_f \boldsymbol{\iota}_{N_I} + \frac{1}{2} \boldsymbol{\sigma}_{i,t}^2 \right].$$

A foreign investor f can invest in the riskless bond, the domestic securities and the foreign securities. Given his investment opportunity set, he solves a similar optimization problem as the domestic investor. The vector $\boldsymbol{\omega}_{f,t}$ of optimal portfolio weights for the foreign unrestricted investor can be written as,

$$\boldsymbol{\omega}_{f,t} = \frac{1}{\gamma^f} \Sigma_t^{-1} E_t \left[\mathbf{r}_{t+1} - \mathbf{c}_{t+1} - r_f \boldsymbol{\iota}_N + \frac{1}{2} \boldsymbol{\sigma}_t^2 \right]$$

where $\boldsymbol{\sigma}_t^2$ is the vector containing the variance of investable and non-investable asset net log-returns i.e. the diagonal elements of Σ_t , γ^f is the coefficient of relative risk aversion of the foreign investor, and $\boldsymbol{\iota}_N$ is an $(N \times 1)$ vector of ones.

Using market clearing conditions and assuming that equities are in net positive supply equal to their market capitalization and the bonds are in zero net supply, we obtain the equilibrium asset pricing relationships.

¹²[Campbell and Viceira \(2002\)](#) use a Taylor approximation of the non-linear function relating log asset returns to log portfolio returns.

2.3 Equilibrium risk and return

2.3.1 Investable set

The investable securities are priced as if the market is fully integrated. The expected net return for an investable security $i \in \{1, \dots, N_I\}$ is given by,

$$E_t [r_{i,t+1}^{net} - r_f] + \frac{1}{2} var_t (r_{i,t+1}^{net}) = \gamma_t cov_t (r_{i,t+1}^{net}, r_{W,t+1}^{net}) \quad (1)$$

where $\gamma_t = \left(\frac{W_t^d}{\gamma^d} + \frac{W_t^f}{\gamma^f} \right)^{-1}$ W_t is the aggregate relative risk aversion coefficient, $W \equiv W^d + W^f$ is the aggregate global wealth and $r_{W,t+1}^{net}$ is the net log-return on the world market portfolio. We then express the model in terms of gross returns,

$$\begin{aligned} E_t [r_{i,t+1} - r_f] &= -\frac{1}{2} var_t (r_{i,t+1}^{net}) + E_t [c_{i,t+1}] + \gamma_t [cov_t (r_{i,t+1}, r_{W,t+1}) \\ &\quad + cov_t (c_{i,t+1}, c_{W,t+1}) - cov_t (c_{i,t+1}, r_{W,t+1}) - cov_t (r_{i,t+1}, c_{W,t+1})]. \end{aligned} \quad (2)$$

The asset pricing equation of investable securities is similar to the liquidity-adjusted CAPM of AP but the market risk premium and liquidity risk premiums are aggregated at the world level and with the addition of one-half the variance of excess returns which obtains when converting from log returns to simple returns. Hence, as in BHL local liquidity variables affect expected returns even under full integration. The asset pricing equation under full market integration of BHL is not identical to ours as BHL assumes that liquidity is a second risk factor that affects the world pricing kernel in addition to the world market risk factor and hence triggers its own price of liquidity risk. Except for the Jensen's inequality term, our pricing equation for investables is the extension of AP to an international setup.

2.3.2 Non-investable set

For a non-investable security $n \in \{1, \dots, N_N\}$, the expected net excess return at equilibrium is given by,

$$E_t [r_{n,t+1}^{net} - r_f] + \frac{1}{2} \text{var}_t (r_{n,t+1}^{net}) = \gamma_t \text{cov}_t (r_{n,t+1}^{net}, r_{W,t+1}^{net}) + \pi_t \text{cov}_t (r_{n,t+1}^{net}, r_{N,t+1}^{net} \mid \mathbf{r}_{i,t+1}^{net}) \quad (3)$$

where $\pi_t = \left(\frac{\gamma_t^f}{W_t^f} - \frac{\gamma_t}{W_t} \right) M_{N,t}$ is the price of local conditional risk. We then express the covariances in terms of gross returns

$$\begin{aligned} \text{cov}_t (r_{n,t+1}^{net}, r_{W,t+1}^{net}) &= \text{cov}_t (r_{n,t+1} - c_{n,t+1}, r_{W,t+1} - c_{W,t+1}) \\ &= \text{cov}_t (r_{n,t+1}, r_{W,t+1}) + \text{cov}_t (c_{n,t+1}, c_{W,t+1}) \\ &\quad - \text{cov}_t (c_{n,t+1}, r_{W,t+1}) - \text{cov}_t (r_{n,t+1}, c_{W,t+1}), \end{aligned} \quad (4)$$

$$\begin{aligned} \text{cov}_t (r_{n,t+1}^{net}, r_{N,t+1}^{net} \mid \mathbf{r}_{i,t+1}^{net}) &= \text{cov}_t (r_{n,t+1}, r_{N,t+1} \mid \mathbf{r}_{i,t+1}^{net}) + \text{cov}_t (c_{n,t+1}, c_{N,t+1} \mid \mathbf{r}_{i,t+1}^{net}) \\ &\quad - \text{cov}_t (c_{n,t+1}, r_{N,t+1} \mid \mathbf{r}_{i,t+1}^{net}) - \text{cov}_t (r_{n,t+1}, c_{N,t+1} \mid \mathbf{r}_{i,t+1}^{net}). \end{aligned} \quad (5)$$

Substituting (4) and (5) in (3) we obtain,

$$\begin{aligned} E_t [r_{n,t+1} - r_f] &= -\frac{1}{2} \text{var}_t (r_{n,t+1}^{net}) + E_t [c_{n,t+1}] + \gamma_t [\text{cov}_t (r_{n,t+1}, r_{W,t+1}) + \text{cov}_t (c_{n,t+1}, c_{W,t+1}) \\ &\quad - \text{cov}_t (c_{n,t+1}, r_{W,t+1}) - \text{cov}_t (r_{n,t+1}, c_{W,t+1})] \\ &\quad + \pi_t [\text{cov}_t (r_{n,t+1}, r_{N,t+1} \mid \mathbf{r}_{i,t+1}^{net}) + \text{cov}_t (c_{n,t+1}, c_{N,t+1} \mid \mathbf{r}_{i,t+1}^{net}) \\ &\quad - \text{cov}_t (c_{n,t+1}, r_{N,t+1} \mid \mathbf{r}_{i,t+1}^{net}) - \text{cov}_t (r_{n,t+1}, c_{N,t+1} \mid \mathbf{r}_{i,t+1}^{net})]. \end{aligned} \quad (6)$$

The extra conditional liquidity risks are

1. $\text{cov}_t (c_{n,t+1}, c_{N,t+1} \mid \mathbf{r}_{i,t+1}^{net})$: the conditional commonality in transaction risk between a security n and the local non-investable market N ;
2. $\text{cov}_t (c_{n,t+1}, r_{N,t+1} \mid \mathbf{r}_{i,t+1}^{net})$: the conditional covariance between a security's transaction cost and the local non-investable market return;

3. $cov_t(r_{n,t+1}, c_{N,t+1} | \mathbf{r}_{i,t+1}^{net})$: the conditional covariance between a security's return and the local non-investable market transaction cost.

As in AP, (1) affects required returns positively because investors want to be compensated for holding securities with higher transaction costs when the local market transaction costs increase; (2) and (3) affect required returns negatively because investors are willing to accept a lower return on an asset with lower transaction costs in a down local market, or an asset with higher return when local market transaction costs are high.

Note that the equilibrium asset pricing model implies that the global risk is measured vis-à-vis the world market return (and aggregate liquidity) as the total world market portfolio is the correct portfolio to measure risk sharing.

To gain further insight into the price of conditional risk, we write the conditional covariance as,

$$cov_t(r_{n,t+1}^{net}, r_{N,t+1}^{net} | \mathbf{r}_{i,t+1}^{net}) = cov_t(r_{n,t+1}^{net}, r_{N,t+1}^{net}) - cov_t(r_{n,t+1}^{net}, r_{DP,t+1}^{net}) \quad (7)$$

where $\mathbf{r}_{DP,t+1}^{net} = \mathbf{r}_{i,t+1}^{net, \top} \Sigma_{ii,t}^{-1} \Sigma_{in,t} \mathbf{M}_{n,t} / M_{N,t}$ is the net return on the diversification portfolio DP , which is the portfolio of investable assets that is most highly correlated with the market portfolio of non-investable securities. The DP portfolio is the one held long by the domestic investor as the best substitute for the non-investable segment, but short by the foreign investor to reduce his local risk exposure.

If the correlation between returns on the value-weighted market portfolio of non-investable securities, $r_{N,t}$, and its diversification portfolio, $\rho_{N,DP,t}$, is one, the market is effectively integrated, all of the conditional market risk premium and conditional liquidity risk premiums disappear and only global market and liquidity risk are priced as in AP. Hence perfect spanning would eliminate the emerging market valuation discount due to illiquidity risk but would not eliminate the global premiums or discounts that result from co-variation of the asset return and its transaction cost with the global market return and global transaction costs. Under complete segmentation, only local market and local liquidity risks are priced. Under partial segmentation, both global and conditional local market return and liquidity risks are priced.

We can capture the degree of segmentation of the foreign market by defining the integration index $II_t = \rho_{N,DP,t}^2$. By aggregating Equation (7) over the set of non-investable securities we

obtain,

$$\text{var}_t(r_{N,t+1}^{net} | \mathbf{r}_{i,t+1}^{net}) = \text{var}_t(r_{N,t+1}^{net}) (1 - II_t). \quad (8)$$

We can then obtain the conditional expected gross return on the local non-investable market portfolio, $r_{N,t+1}$ as,

$$\begin{aligned} E_t[r_{N,t+1} - r_f] &= -\frac{1}{2}\text{var}_t(r_{N,t+1}^{net}) + E_t[c_{N,t+1}] + \gamma_t[\text{cov}_t(r_{N,t+1}, r_{W,t+1}) + \text{cov}_t(c_{N,t+1}, c_{W,t+1}) \\ &\quad - \text{cov}_t(c_{N,t+1}, r_{W,t+1}) - \text{cov}_t(r_{N,t+1}, c_{W,t+1})] \\ &\quad + \pi_t[\text{var}_t(r_{N,t+1}) + \text{var}_t(c_{N,t+1}) \\ &\quad - 2\text{cov}_t(r_{N,t+1}, c_{N,t+1})] (1 - II_t). \end{aligned} \quad (9)$$

which is the fundamental asset pricing equation that we will use in the empirical analysis. It is useful to write the model in gross returns to examine the role of liquidity risk premiums. However, we cannot estimate the model in a conditional setup using gross returns because the local covariance is also conditional on net returns. Hence, to test the asset pricing model, we use the conditional asset pricing model in net returns as in equation 3. We then compute the different covariance terms to investigate the importance of global and liquidity risk for measuring risk premia.

3 Econometric specification

Our empirical methodology follows three main steps. In a first step, we estimate univariate GARCH models and copula models to obtain, respectively, variances and covariances needed to estimate the time-varying weights of the diversification portfolio. In a second step, we estimate the time-varying price of world market risk and the coefficients of the time-varying world variance. In the third step, we impose these estimates in the country estimations. For each country, we have a system of three equations that include the world market portfolio, the non-investable stock market portfolio, and its diversification portfolio. We then use the multivariate generalized autoregressive conditional heteroskedasticity (GARCH)-in-mean methodology with a diagonal covariance specification (see [Bollerslev, Engle, and Wooldridge \(1988\)](#)). The estimation procedure results in sampling errors from the two first stages. However, imposing the world price of risk in the third step is in line with the theoretical model restrictions and produces more powerful tests.

3.1 Estimation of the diversification portfolio

The diversification portfolio for country k is composed of the value-weighted market portfolio W , the value-weighted portfolio of investable securities I^k , the value-weighted portfolio of substitute assets S^k which includes ADRs, GDRs, and direct listings, and all its Country Funds and related Exchange Traded Funds, which we denote as $CF^k = \{CF_1^k, CF_2^k, \dots\}$. We discuss in Section 4 our construction methodology for each of these portfolios. We form a dynamic replicating portfolio whose allocation at time t uses conditional covariances:

$$\omega_{DP^k,t} = \Sigma_{(W,I^k,S^k,CF^k),t}^{-1} \Sigma_{N^k-(W,I^k,S^k,CF^k),t} \quad (10)$$

where $\Sigma_{(W,I^k,S^k,CF^k),t}$ is the covariance matrix at time t of the vector

$$\mathbf{r}_{(W,I^k,S^k,CF^k),t} = \left[r_{W,t}, r_{I^k,t}, r_{S^k,t}, r_{CF_1^k,t}, r_{CF_2^k,t}, \dots \right]^\top$$

and $\Sigma_{N^k-(W,I^k,S^k,CF^k),t}$ is the vector of covariances between the return on the value-weighted portfolio of non-investable securities $r_{N^k,t}$ and the vector $\mathbf{r}_{(W,I^k,S^k,CF^k),t}$.

We estimate one volatility model for each asset and then obtain the correlations from a dynamic normal copula (DNC) estimated on the residuals as in [Christoffersen and Langlois \(2013\)](#). For each asset, we estimate an AR(1)-NGARCH model on net returns,

$$\begin{aligned} r_{j,t+1} - c_{j,t+1} &= \phi_{j,0} + \phi_{j,1}(r_{j,t} - c_{j,t}) + \epsilon_{j,t+1}, \quad \epsilon_{j,t+1} \sim N(0, \sigma_{j,t+1}^2) \\ \sigma_{j,t+1}^2 &= \sigma_{0,j}^2 (1 - b_j - a_j(1 + c_j^2)) + b_j \sigma_{j,t}^2 + a_j (\epsilon_{j,t} - \sigma_{j,t} c_j)^2 \end{aligned}$$

where $j \in \{W, N^k, I^k, S^k, CF_1^k, CF_2^k, \dots\}$ and $b_j + a_j(1 + c_j^2) < 1$. Then, we estimate one DNC model in which the dynamic correlations are driven by the dynamic,

$$\Gamma_{t+1} = \Omega_C (1 - b_C - a_C) + b_C \Gamma_t + a_C u_t u_t^\top$$

where Ω_C is the long-run correlation matrix, b_C and a_C are the correlation dynamic parameters with $b_C + a_C < 1$, and $u_{j,t} = \sqrt{\Gamma_{j,j,t}} \frac{\epsilon_{j,t}}{\sigma_{j,t}}$ are the standardized residuals with the [Aielli \(2013\)](#)

correction. We obtain the dynamic correlation through the standardization $C_{i,j,t} = \frac{\Gamma_{i,j,t}}{\sqrt{\Gamma_{i,i,t}\Gamma_{j,j,t}}}$.

Estimating conditional covariances is complicated by the fact that the replicating assets have different time series length, with some series not overlapping at all. To ensure positive definiteness of the covariance matrix of $r_{N^k,t}$ and $r_{(W,I^k,S^k,CF^k),t}$, we estimate the DNC model by maximizing the log-likelihood using all available assets at each time period. This allows us to estimate one correlation model on a unbalanced panel of asset returns. Once an asset k becomes available, we set its first conditional correlations with other available assets l to its long-run parameter $\Omega_{k,l}$. We examine the unbiasedness of the estimators via simulations. The results suggest that the estimator of the DNC with missing values in the panel is unbiased.

Since the distribution of the GARCH shocks $\epsilon_{j,t}$ and the copula is gaussian, the joint distribution obtained by combining them is also multivariate gaussian. Hence, we use the conditional variances obtained from the AR(1)-NGARCH models and the conditional correlations obtained from the DNC model to obtain the conditional covariances needed to estimate the diversification portfolio weights.

3.2 Asset pricing tests

We test the conditional asset pricing model for each country k . Specifically, we estimate the following system of equations,

$$\begin{aligned} r_{W,t} - c_{W,t} - r_{f,t} + \frac{1}{2}\sigma_{W,t}^2 &= \alpha_W + \kappa_W E[c_{W,t}] + \gamma_{t-1}\sigma_{W,t}^2 + \epsilon_{W,t} \\ r_{N^k,t} - c_{N^k,t} - r_{f,t} + \frac{1}{2}\sigma_{N^k,t}^2 &= \alpha_{N^k} + \kappa_{N^k} E[c_{N^k,t}] + \gamma_{t-1}\Sigma_{N^k,W,t} \\ &\quad + \pi_{t-1}^k \sigma_{N^k,t}^2 \left(1 - \frac{\Sigma_{N^k,DP^k,t}^2}{\sigma_{N^k,t}^2 \sigma_{DP^k,t}^2}\right) + \epsilon_{N^k,t} \\ r_{DP^k,t} - c_{DP^k,t} - r_{f,t} + \frac{1}{2}\sigma_{DP^k,t}^2 &= \alpha_{DP^k} + \kappa_{DP^k} E[c_{DP^k,t}] + \gamma_{t-1}\Sigma_{DP^k,W,t} + \epsilon_{DP^k,t} \end{aligned} \quad (11)$$

where $\Sigma_{i,j,t}$ are the elements of the (3×3) covariance matrix Σ_t of the asset net returns. We allow for intercepts $\alpha_j, j \in \{W, N^K, DP^K\}$ in the estimation but the model implies that the intercepts are zero.

In the next section, we use the bid-ask spread measure of [Corwin and Schultz \(2012\)](#) estimated from the high-low price spread. To test our model, we estimate three specifications. First, we impose

the model-implied constraint that $\kappa_W = \kappa_{N^k} = \kappa_{DP^k} = 0$ and use it as our main specification. Admittedly excess returns net of bid-ask spreads overestimate the true cost of trading. The asset pricing model assumes that all investors have the same horizon, an assumption unlikely to be verified in reality. In a second specification discussed in Section 6, we follow AP and include on the right hand side the expected transaction cost to capture the fact that the effect of expected transaction costs in the data may differ from our stylized model. In a third specification, we estimate the model in gross returns where the liquidity costs are set to zero $c_{j,t} = 0, \forall j, t$. This is the standard EL asset pricing model that we use as the benchmark specification.

We allow the price of world market risk to change through time as suggested in the literature (see, among others, [Harvey \(1991\)](#) and [Santis and Gerard \(1997\)](#)). In order to estimate the model, we need to specify the functional form of the time-varying price of world covariance risk. Because the model implies that the price of world market risk should be positive, we model its dynamics using a square function of the global instruments $\mathbf{Z}_{W,t-1} = (1, \mathbf{Z}_{W,t-1}^{*\top})^\top$,

$$\gamma_t = (\lambda_W^\top \mathbf{Z}_{W,t-1})^2. \quad (12)$$

If the world market risk is priced, we should reject the hypothesis that the $\lambda_{i,W} = 0$, for $i \geq 1$. If the world market risk is time-varying, we should reject that the $\lambda_{i,W} = 0$, for $i > 1$.

Similarly, we allow the price of conditional market risk to be time-varying. Because the price of conditional market risk should be positive, we model the price dynamics using a square function of the local instruments $\mathbf{Z}_{k,t-1} = (1, \mathbf{Z}_{k,t-1}^{*\top})^\top$,

$$\pi_t^k = (\lambda_k^\top \mathbf{Z}_{k,t-1})^2.$$

If local risk is priced, we should reject the hypothesis that the $\lambda_{i,k} = 0$, for $i \geq 1$. If local risk is time-varying, we should reject the hypothesis that the $\lambda_{i,k} = 0$, for $i > 1$.

The choice of instruments used in the parametrization of the prices of risk follow the literature (See among others [Ferson and Harvey \(1993\)](#), [Ferson and Harvey \(1994\)](#), [Dumas and Solnik \(1995\)](#), [Santis and Gerard \(1997\)](#)). The global instruments $\mathbf{Z}_{W,t-1}$ include the world dividend yield in excess of the risk free rate, the change in the U.S. term premium, measured by the yield difference

between the three-month T-bill and the 10-year T-bond, and the U.S. default premium measured by the yield difference between Moody's Baa and Aaa rated bonds. The local instruments $\mathbf{Z}_{k,t-1}$ include the local equity return in excess of the risk free rate, the local excess dividend yield and the change in local nominal exchange rate.

We use the multivariate generalized autoregressive conditional heteroskedasticity (GARCH)-in-mean methodology with a diagonal covariance specification (see [Bollerslev et al. \(1988\)](#)), which assumes that the variances depend only on past squared residuals and an autoregressive component, while the covariances depend on the past cross-product of residuals and an autoregressive component. The dynamics of the conditional second moment Σ_t are specified as,

$$\Sigma_t = \Sigma_0 \circ (\boldsymbol{\iota}\boldsymbol{\iota}^\top - \mathbf{b}\mathbf{b}^\top - \mathbf{a}\mathbf{a}^\top) + \mathbf{b}\mathbf{b}^\top \circ \Sigma_{t-1} + \mathbf{a}\mathbf{a}^\top \circ \boldsymbol{\epsilon}_{t-1}\boldsymbol{\epsilon}_{t-1}^\top \quad (13)$$

where $\boldsymbol{\iota}$ is a (3×1) vector of ones, $\mathbf{a}$ and $\mathbf{b}$ are (3×1) vectors of unknown parameters, $\circ$ is the Hadamard (element by element) matrix product, and each element of $\mathbf{b}\mathbf{b}^\top + \mathbf{a}\mathbf{a}^\top$ is smaller than one. The advantage of this multivariate GARCH in mean parameterization is that it ensures positive definiteness of the covariance matrix while reducing the number of parameters to be estimated.

Under normality assumption, the log likelihood function can be written as,

$$\ln L(\boldsymbol{\Theta}) = -\frac{TN}{2} \ln(2\pi) - \frac{1}{2} \sum_t^T \ln(\det(\Sigma_t(\boldsymbol{\Theta}))) + \boldsymbol{\epsilon}_t^\top(\boldsymbol{\Theta}) \Sigma_t^{-1}(\boldsymbol{\Theta}) \boldsymbol{\epsilon}_t(\boldsymbol{\Theta}) \quad (14)$$

where $\boldsymbol{\Theta}$ is the vector of parameters and $\boldsymbol{\epsilon}_t(\boldsymbol{\Theta})$ is the (3×1) vector of innovations. Since normality is often violated, we estimate the model and compute the parameters' covariance matrix by the quasi-maximum likelihood (QML) of [Bollerslev and Wooldridge \(1992\)](#), using the BFGS ([Shanno \(1985\)](#)) algorithm to update the Hessian.

Since the theory predicts that the world price of risk should be the same for each country, we first estimate the world equation in the system of equations (11) to obtain the time-varying world price of risk and the coefficients of the time-varying world variance (see, for example, [Bekaert and Harvey \(1995\)](#)). We then impose the estimates $\hat{\kappa}_W, \hat{E}_{t-1}[c_{W,t}], \hat{\gamma}_{t-1}, \hat{\sigma}_{W,0}^2, \hat{\alpha}_W$, and $\hat{\beta}_W$ in the country estimations.

4 Data

We include all emerging markets that were at some point in the MSCI Emerging Market index (Argentina, Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Israel, Jordan, Malaysia, Mexico, Morocco, Pakistan, Peru, Philippines, Poland, Portugal, Qatar, Russian Federation, South Africa, South Korea, Sri Lanka, Taiwan, Thailand, Turkey, United Arab Emirates, and Venezuela). For each of these markets, we obtain from Datastream daily low, high, and closing prices, trading volumes, total returns, and monthly market capitalizations in both local and U.S. currencies for all securities classified as equity, American Depositary Receipts, and Global Depositary Receipts. To test our asset pricing model, we use daily data to build our monthly liquidity measures and U.S. dollar denominated returns in excess of the one-month Treasury bill rate obtained from Kenneth French’s website. We use the longest times series available for each security, and our sample ends in December 2015.

We apply the filters on daily and monthly data listed in Appendix A. As pointed out in [Ince and Porter \(2006\)](#), Datastream data has to be handled with care, and we follow [Lee \(2011\)](#) and [Karolyi et al. \(2012\)](#) in applying filters to the data.

To test our asset pricing model, we need direct measures of transaction costs. For all stocks, we use the bid-ask spread measure of [Corwin and Schultz \(2012\)](#) to approximate transaction costs. Using two consecutive days, [Corwin and Schultz \(2012\)](#) calculate both daily high-low price spreads and the high-low price spread over the two days to distinguish the bid-ask spread from daily variance. By applying their methodology, we obtain daily estimates of the bid-ask spread. Following their suggestion, we set negative daily spreads to zero. The monthly spread measure is the average of daily measures within the month for months in which there are at least ten daily spreads available. While it is a good proxy for the cost of a round-trip transaction of small size, this measure does not capture other dimensions of liquidity such as market depth, resilience and search costs.

The non-investable and investable portfolios for each country contain only local stocks. We define a local stock as one traded on the major stock exchange in each country, which is the one with the highest number of equities listed. Exceptions to this rule are Brazil (Rio de Janeiro and Bovespa), China (Hong Kong, Shanghai, and Shenzhen), Germany (Frankfurt and XETRA), Japan (Osaka and Tokyo), Russia (Moscow and MICEX), South Korea (KOSDAQ and Korea

Stock Exchange), and United States (NASDAQ, NYSE, and NYSE Arca). We test our asset pricing model on emerging markets, but use stock exchanges from developed markets to determine Depository Receipts' (DR) and direct listings' eligibility and to build the world market portfolio as discussed below. To ensure we have local common stocks, we apply the filters from [Griffin, Kelly, and Nardari \(2010\)](#) on local stocks' names and industries.

Each month, we build value-weighted return and bid-ask spread indices. We construct one set of indices that contains all stocks for a given country, one that contains only investable stocks, and one that contains only non-investable stocks. The compositions of our portfolios change every month. To be eligible, a stock must have over the last three months valid U.S. dollar denominated monthly returns, market capitalizations, and bid-ask spreads. Then, we follow [Karolyi and Wu \(2012\)](#) and characterize a stock as investable if there exists a related cross-listed stock, an ADR, or a GDR traded on a major stock exchange in a developed country with a valid monthly return. We identify cross-listing stocks using ISINs. We identify ADRs and GDRs by retrieving from Bloomberg all their underlying SEDOLs and ISINs¹³. We also use country specific filters; B shares in China and the Philippines are classified as investable.

Alternative determinants of investability include the investability weight factor (IWF) computed by S&P/IFC or the foreign inclusion factor (FIF) computed by MSCI. As a robustness test discussed in Section 6, we use the IWF at the stock level to separate investables and non-investables¹⁴. Another investability proxy is the foreign ownership holdings used in [Errunza and Ta \(2015\)](#). The foreign ownership holdings data from Lionshare start in 1999 and do not change the sample of investable stocks compared to the existence of depository receipts or direct listings as determinant of the investability of a stock as in [Karolyi and Wu \(2012\)](#). [Carrieri, Chaieb, and Errunza \(2013\)](#) show that implicit barriers to foreign investments results in investable stocks not effectively integrated with the world economy. Therefore, we use in Section 6 the portfolio of all local stocks as our non-investable set.

We use all local stocks listed in 53 developed and emerging markets to build the value-weighted world market portfolio. For each country, we build a portfolio of all eligible ADRs, GDRs, and direct listings. The objective is to obtain the best replicating asset for the portfolio of local non-

¹³We are able to match 2,950 of the 3,729 DRs using Bloomberg

¹⁴The FIFs are not widely available before 2001

investable stocks. Hence, for each stock for which there exists a DR or a direct listing, we form a value-weighted portfolio of all DRs and direct listings using their respective market capitalizations. The weight of each of these portfolios in the portfolio of substitute assets S^k is proportional to the market capitalization of their respective local stock. Hence, we obtain a portfolio that is as close as possible to the non-investable portfolio of local stocks and that allocates more to larger DRs and direct listings. We also use a list of Exchange Traded Funds (ETF) and country funds (CF) for each country mainly compiled from Morningstar, Bloomberg, iShares and CRSP. These assets will be used to replicate the portfolio of non-investable assets. Note that when we form the diversification portfolio, we use as replicating assets the world market portfolio W , the portfolio of investable stocks I^k , the portfolio of ADRs, GDRs, and direct listings S^k , and all available country funds and ETFs $CF_1^k, CF_2^k, \dots$. When we use in Section 6 as a robustness check the portfolio of all local stocks as the set of non-investable securities, we remove the portfolio of investable stocks from the set of replicating assets.

To be able to estimate the dynamics of variances and covariances, we only keep countries for which we have at least a 10-year time series for both the non-investable and the investable index. Our final sample is comprised of the following 21 countries: Argentina, Brazil, Chile, China, Egypt, Greece, Hungary, India, Indonesia, Israel, Malaysia, Mexico, Morocco, Philippines, Poland, Russia, South Africa, South Korea, Taiwan, Thailand and Turkey. Further, we require at least a 10-year time series for replicating assets to be included in the diversification portfolio.

4.1 Summary statistics

Table 1 reports the number of stocks in the sample every five years from 1988-2015. The table also reports the descriptive statistics of the monthly returns and the [Corwin and Schultz \(2012\)](#) bid-ask spread measure.

The number of stocks varies across countries and years as a result of data coverage and the set of filters detailed in the Appendix A. It is quite small at the beginning of the period but increases steadily. Table 1 also shows the fraction of investables reported in column (1) in italics. There are 642 stocks in 1988 of which 1% are investable and 8,686 stocks at the end of the sample in December 2015 with 6% investable. The proportion of investable stocks steadily increases until the mid-2000s, but has stagnated since. In 2015, China is the country with the largest number of

stocks in the sample with 2,219 stocks of which 183 are investables, while Morocco has the smallest number of stocks with only 14 stocks that are essentially non-investable. Hungary and Mexico are the countries with the largest fraction of investables standing at 47% (9 over 19) and 51% (39 over 77), respectively. In the country estimations, we use the longest sample period in which the non-investable portfolio is available, and therefore the starting date varies across countries.

The last four columns of Table 1 show averages and volatility of the monthly excess returns and of the bid-ask spread measure. The return figures are annualized and for comparison purposes the statistics are computed over the sample period during which both the non-investable and the investable index are available. The U.S. dollar monthly returns are in excess of the 1-month U.S. T-bill rate. The emerging markets display high average returns but also high volatility as well-documented in the literature. Annualized equity volatilities range between 16% and 63%.

[Insert Table 1]

We observe striking cross-country differences in the value-weighted average illiquidity of the emerging market indexes. According to the bid-ask spread measure, India is on average the least liquid, while Chile is the most liquid. We report in italics in column (11) of Table 1, the ratio of the average bid-ask spread for non-investables to the average bid-ask spread for investables. The ratio is above one for all but three countries and it reaches 1.88 for Russia, which indicates that non-investables are more illiquid than investables. To further examine the difference in liquidity measures and their dynamics across investables and non-investables, we report in Figure 1 value- and equal-weighted averages of bid-ask spread measures across countries for portfolios of non-investable stocks (left column) and investable stocks (right column). In each graph, the horizontal line reports the time-series average. The plots show interesting differences across investables and non-investables. Investables are most of the time and on average more liquid than non-investables. There is also no clear downward trend over time in bid-ask spreads for both investables and non-investables. The plots also show that illiquidity varied significantly over time. The impact of the Asian, the Russian default, the Long Term Capital Management (LTCM), and the subprime crises on bid-ask spreads is clear. The rise in illiquidity during the 2008-2009 crisis is quite strong for both investables and non-investables. After the global financial crisis, illiquidity reverted back to average levels for both investables and non-investables.

[Insert Figure 1]

5 Results

In this section, we estimate our asset pricing model and test whether world market and conditional market risk factors are priced and are time-varying. We also examine the economic contribution and the dynamics of the different risk premia, specifically the global and local liquidity premia. We do this by running the multivariate GARCH-in-mean on the world market portfolio, the EM non-investables and their diversification portfolios as detailed in Section 3.

5.1 Prices of risk

First we estimate the price of world market risk in a conditional set up allowing the price to vary over time ($Z_{W,t} = (\mathbf{1}, Z_{W,t}^*)'$). We test the world market model using the three model specifications; (1) Main: net returns imposing a zero κ , (2) Alternative: net returns and unconstrained κ , and (3) Benchmark: gross returns obtained by setting the illiquidity cost to zero in the system of equations 11. Panel A of Table 2 shows that the price of world market risk is statistically significant and time-varying at the 1% level, and has an average value of two over the three model specifications. The price of world market risk is proportional to the aggregate risk aversion. Hence, a value of two is economically meaningful. Figure 2 reports the time varying price of world market risk. The price of world market risk increases during economic contractions and crises periods.¹⁵ Additionally, the model estimation from net returns and unconstrained kappa results in an estimate of aggregate risk aversion that is higher and more economically plausible compared to the estimate from gross returns and from net returns with zero κ . The average estimates from the three specifications are 1.25, 3.3 and 1.45 for the main, alternative and benchmark specifications, respectively. While this is an interesting finding, the intercepts from the world market equation are insignificant for the main and benchmark specifications but is significant for the alternative one. For the local prices of risk we report and discuss the main and benchmark specifications. The results from the alternative specification are discussed in Section 6.

¹⁵The period of contraction is measured from peak to trough as determined by the National Bureau of Economic Research, NBER. The crises periods include the Asian, Tequila, LTCM, and global financial crisis.

[Insert Figure 2]

Figure 3 plots the median, 25%, and 75% of the local prices of risk for our EM sample obtained from the third step of estimation and our main specification. The prices fluctuate through time and across countries and are quite volatile. They tend to increase during economic recessions and financial crises. The figure also reports the average local price across the 21 countries and across the sample period. The cross-country average price of local risk is about five. The price of local risk is proportional to the differential in risk aversion between an EM average investor and the world average investor and hence the size seems economically plausible.

[Insert Figure 3]

Panel B of Table 2 contains a number of specification tests. For each country, we report robust Wald tests for the significance and time variation in the prices of conditional local market risk in columns (2)-(5) ¹⁶. It is important to note that the p-values we report ignore the sampling error associated with two first step estimations. Our main specification shows that the price of local market risk is significant and time varying for the sample of EM countries. The model estimation with gross returns confirm these results. Panel B of Table 2 also shows the intercepts of the market portfolio of noninvestables and its DP for the main and benchmark specifications in columns (6)-(9). The intercept from the main specification is significant in only two of 21 countries, namely, Chile and Mexico. Similarly the estimation with gross returns shows a non zero intercept for only Mexico. Overall, the results lend support to both models. Notice that both the main and benchmark specifications have the same number of factors and hence same degrees of freedom. The only difference is the liquidity adjustment. Nevertheless we cannot compare the pseudo R^2 from the two specifications as the dependent variable is different across the two models.

We do not report the parameter estimates ($\mathbf{a}$ and $\mathbf{b}$) for the conditional covariance processes, but these are significant in all cases, of the usual size, and satisfy the stationarity condition, i.e., each element of $\mathbf{bb}^\top + \mathbf{aa}^\top$ is smaller than one. This value is also close to one in most instances, suggesting that the variance and covariance processes in Σ_t are highly persistent.

[Insert Table 2]

¹⁶The unreported estimates on coefficients used to parameterize the prices of risk are statistically significant in several cases. The results are available from the authors.

5.2 Liquidity risk premium

Next we examine the economic contribution of the world and local liquidity risk premia. As shown in Equation (9), the world liquidity risk premium has three components: the sensitivity of the non-investable returns to the world market-wide illiquidity; the sensitivity of the non-investable illiquidity to the world market return; and the global commonality in illiquidity. The local liquidity risk premium has two components; the conditional variance of non-investables illiquidity and the conditional covariance between non-investables index return and its illiquidity. We obtain the conditional variance and covariances by estimating a bivariate GARCH model. For each country, we estimate the monthly relative contribution of world liquidity risk premium to the world risk premium by computing for each month, the ratio of each world liquidity component to the covariance between non-investable net returns and world market net returns. For each country, we also estimate the monthly relative contribution of local liquidity risk premium to the local risk premium by computing for each month, the ratio of each local liquidity component to the conditional variance of non-investable net returns. Figures 4 and 5 plots at each time t the median across 21 countries of relative contributions of global liquidity risk premium and of local liquidity risk premium, respectively.

[Insert Figures 4 and 5]

The average across countries and time of the contributions of the three global liquidity risk premia amount to 1.16%, 0.5%, and 0.02%, respectively. Over most of the sample period, the sensitivity of non-investable returns to aggregate world illiquidity is the dominant premium, while the global commonality is the least significant among the three. The contribution of the world liquidity risk premiums varies over time. Specifically, the contribution of the sensitivity of non-investables returns to world market illiquidity costs is quite significant during financial crises characterized by significant dry up of liquidity. This is the case in particular during the Tequila crisis, the Russian default and LTCM crisis, the global financial crisis, and the Euro sovereign debt crisis. It is interesting to note that, unconditionally as well as conditional on the state of the market, the covariation of non-investable returns to world market illiquidity is the dominant factor.

The averages across countries and time of the contribution of the two local liquidity risk premia amount to 0.03% and 0.72%, respectively. The conditional covariance between non-investable

returns and their illiquidity is the dominant local liquidity premium. The contribution of local liquidity premium is higher in crises periods but is still largely dominated by the local market risk premium. The average contribution of the local liquidity risk premium to the local risk premium of 0.75% is of similar magnitude to the contribution of the world liquidity risk premium to world risk premium which amounts to 1.68%. To summarize, consistent with past findings, both the world and local liquidity risk premia are quite small on average.

6 Robustness

We run different robustness checks on (1) the model specification, (2) the choice of the non-investable test assets, and (3) the choice of the investability criteria.

6.1 Alternative model specification

The asset pricing model assumes that all investors have the same horizon, an assumption unlikely to be verified in reality. We then follow AP and include on the right hand side the expected transaction cost, which we model as an AR(2) process, to capture the fact that the effect of expected transaction costs in the data may differ from our stylized model. The specification tests in Panel A of Table 3 confirm our previous findings that the local risk is significantly priced and is time varying in all cases. However, the intercepts of the non-investables and/or their corresponding DPs are significant in 7 out of 21 countries. We run a likelihood ratio test that compares the main specification with zero κ and the alternative one with unconstrained κ . That is, we test the restrictions whether the coefficients on both expected illiquidity cost on the market portfolio of non-investables and its DP are jointly equal to zero. In 8 countries out of 21 we reject the zero κ restriction. Also the estimated κ are negative and significant in 14 cases. These results do not lend strong support to the alternative model specification. Nevertheless, the statistical and economic significance of the illiquidity risk premiums from the alternative specification remain unchanged. [The economic contribution obtained with the alternative model specification is available from authors]

[Insert Table 3]

6.2 Choice of the non-investable test assets

Past studies (see, for example, [Carrieri et al., 2013](#)) suggest that although investables are freely available to non-local investor they still command a local risk premium ¹⁷. We then assume that the whole stock market of the emerging country is non-investable. We form the diversification portfolios for the local market portfolio that includes both investables and non-investables and run a system of two equations that include the local market portfolio and its diversification portfolio. Panel B of Table 3 summarizes the hypotheses tests of significant and time-varying conditional local price of risk from the main and benchmark specifications and reports the intercepts of the market portfolio of non-investables and its DP from both model specifications. The local price of risk is significant and time-varying in all cases and across the two specifications. The intercepts from the main specification are insignificant except for India, Mexico, and Taiwan. The intercepts from the benchmark specification are insignificant except for Greece and Mexico. The relative contribution of the liquidity risk premiums are unreported and available from the authors. The magnitudes are similar to the estimation with investables and non-investables and confirm that illiquidity risk premium is valued more in times of financial market stress. All of these findings lend strong support to our liquidity asset pricing model in partially segmented markets.

6.3 Choice of the investability criteria

We examine an alternative determinant of investability based on the S&P/IFC investability screens which include minimum requirements for size, free float, liquidity, foreign inclusion, and length of trading (see S&P/IFC Index Methodology). [TO COMPLETE]

7 Conclusion

We develop a formal international asset pricing model that takes into account cross-border investment barriers as well as random transaction costs to analyze the effects of illiquidity cost and systematic liquidity risk factors on the pricing of EM securities. In our model, the freely traded securities command a global market risk premium and world liquidity premia whereas the securities

¹⁷[Carrieri, Errunza, and Sarkissian \(2004\)](#) show that local factors are priced not only for emerging markets but also for developed markets.

that can be held by only a subset of investors command additionally a conditional local market risk premium and conditional local liquidity risk premia.

We estimate the model for 21 EMs. Empirical test results support theoretical predictions. Specifically, the price of world market risk is positive, statistically and economically significant. The cross-country average price of local risk is about five, however, the magnitude of the price of local risk varies across countries. Although, on average, both the world and local liquidity risk premia are quite small their importance increases during crisis periods corroborating past evidence. Our results are not sensitive to alternative model specification with unrestricted expected liquidity costs, the choice of the set of non-investables, and the investability criteria.

In conclusion, our model provides a formal framework for testing important issues such as liquidity levels and risks in a realistic world market setting and for examining the interaction between investability and illiquidity.

Appendix A - Data filters and corrections

We apply different filters on Datastream daily and monthly data:

- All datatypes do not have the same time series length on Datastream. For all datatypes, we keep only the sample period for which the unadjusted price of the security is available.
- Following [Lee \(2011\)](#) and [Lesmond \(2005\)](#), we drop days for which more than 90% of local stocks in a given country have zero returns on that day.
- When the return over two consecutive days is smaller than 50%, but one of the daily returns is larger than 100%, we remove both daily returns.
- Similarly, when the return over two consecutive months is smaller than 50%, but one of the monthly returns is larger than 300%, we remove both monthly returns.
- To ensure that our monthly returns are not impacted by rounding, we specify the number of decimals to 9 when converting the total return indexes to U.S. dollar using the function $DPL\#(X(RI) \text{ U}\$, 9)$.
- Following [Corwin and Schultz \(2012\)](#), we remove daily data for which:
 1. the ratio of high to low price is smaller than 1 or larger than 8,
 2. any of the low, high, and closing price both for the day and the previous day are unavailable.

We remove monthly bid-ask spread measure larger than 100%.

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Figures and Tables

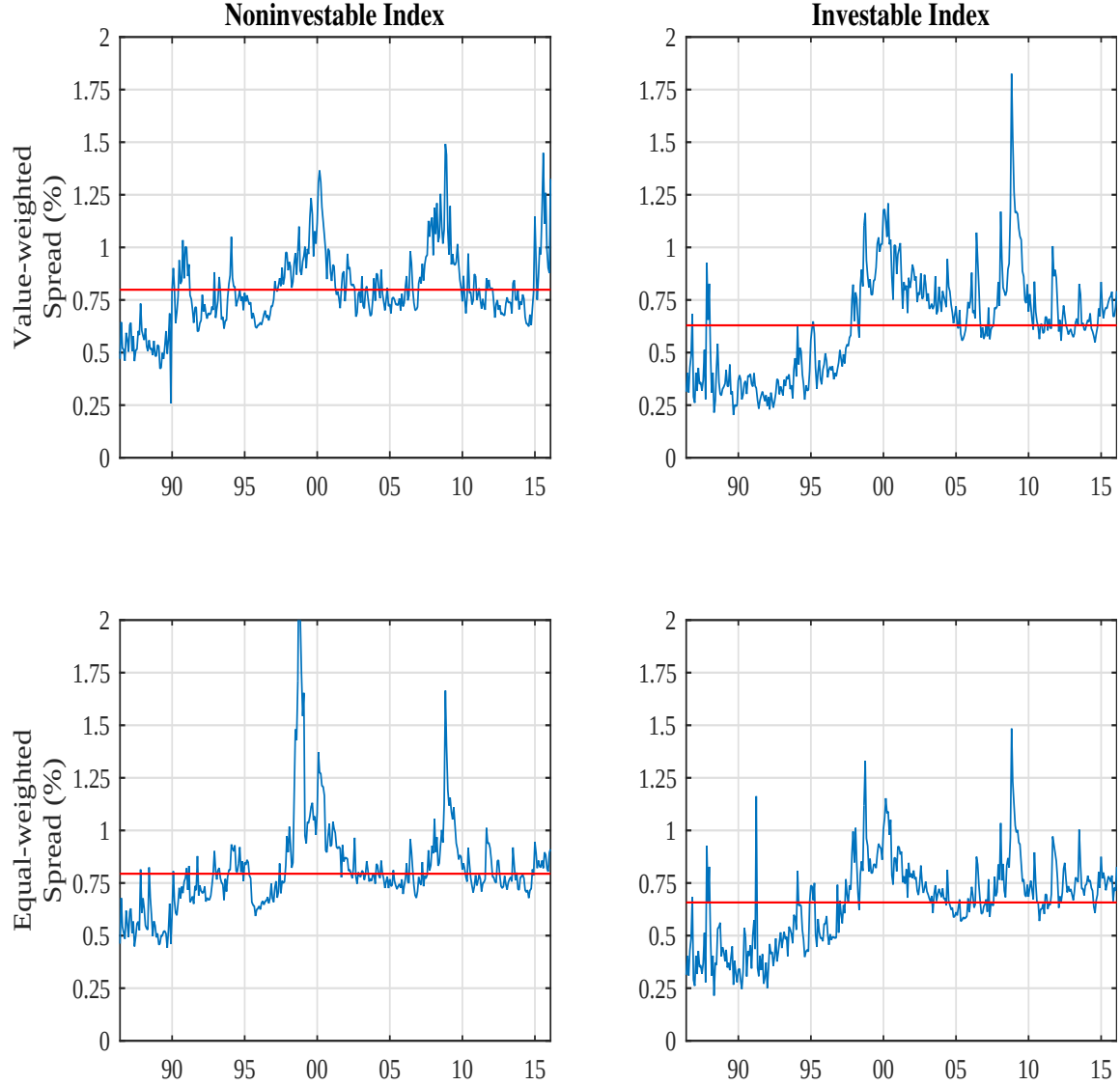


Figure 1 Bid-ask spread for investable and non-investable securities averaged across emerging markets

We report the monthly bid-ask spreads averaged across all 21 emerging markets. For each country, we build value-weighted indices that contain either non-investable or investable stocks. Then, we build cross-country averages of these portfolios' bid-ask spreads. The top row reports on the market capitalization weighted country average, and the bottom row shows equal-weighted country averages. The left column reports the bid-ask spread measures for portfolios of non-investable stocks and the right column for portfolios of investable stocks. In each graph, the horizontal line reports the time-series average of the conditional bid-ask spread measure. We use the high and low price based bid-ask spread measure of [Corwin and Schultz \(2012\)](#).

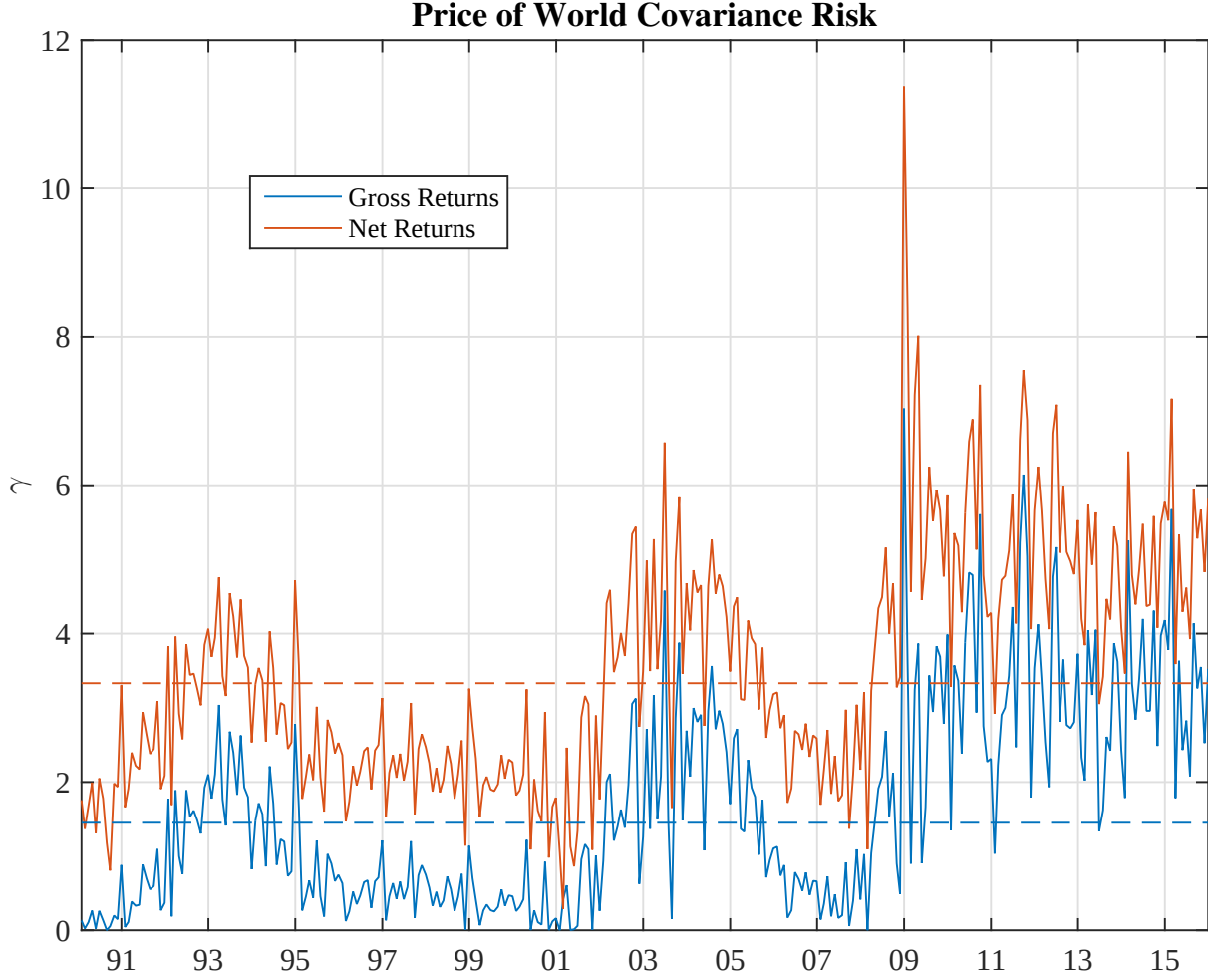


Figure 2 Price of World Covariance Risk γ

We report the conditional price of world covariance risk, γ_t , estimated on gross returns and on net returns. The horizontal dashed lines reports the time-series averages of the conditional prices. The prices of risk are estimated by the methodology of Section 3.2 applied to gross and net returns with unconstrained κ . The sample period is monthly from January 1988 to December 2015.

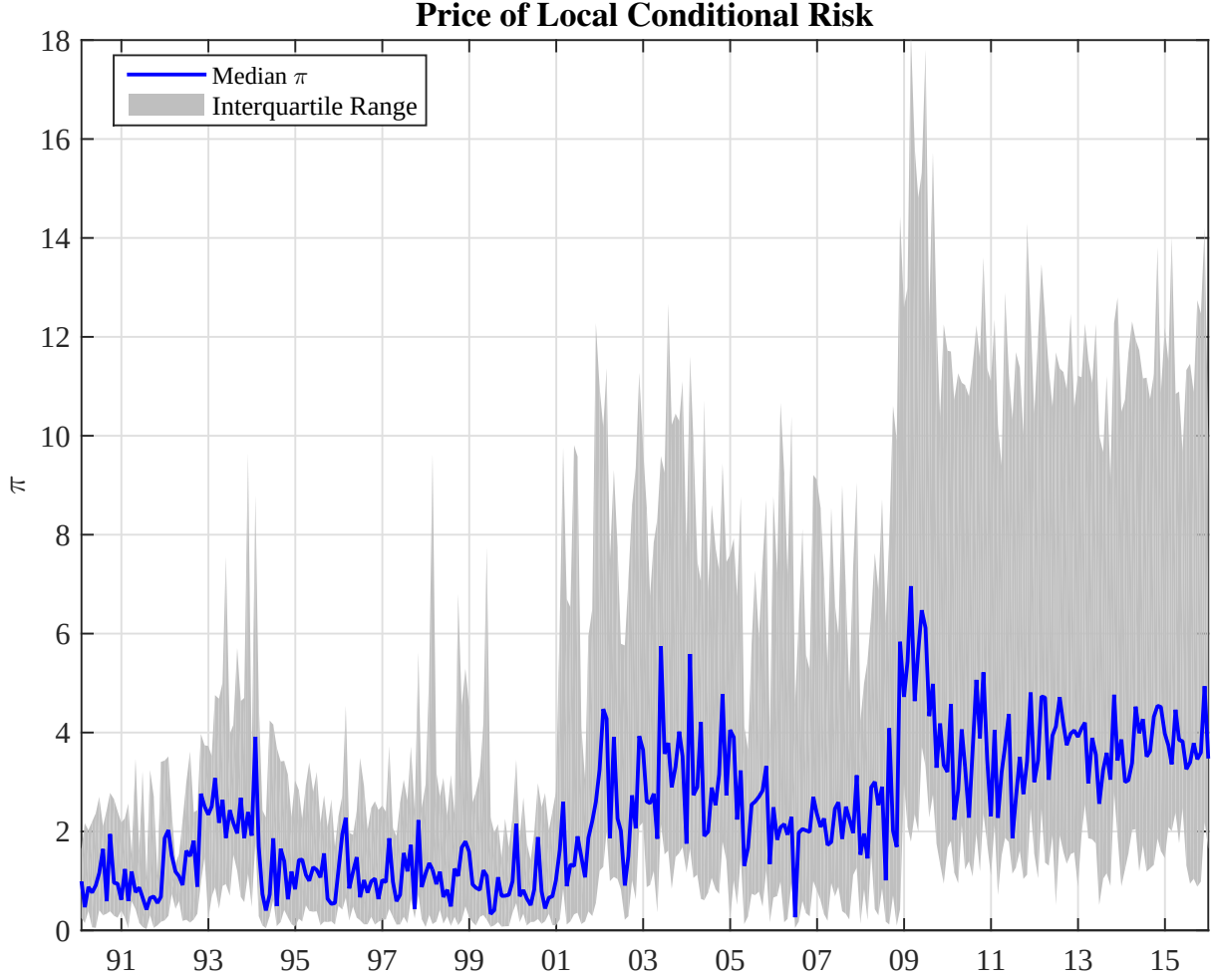


Figure 3 Price of Local Covariance Risk π

We report the time-varying median price of conditional covariance risk, π_t , estimated on net returns. The gray area reports the 25th to 75th range of prices computed across all available countries at each point in time. The horizontal dashed line is the average price across countries and over time. The prices of risk are estimated by the methodology of Section 3.2 applied to net returns. See equation system 11.

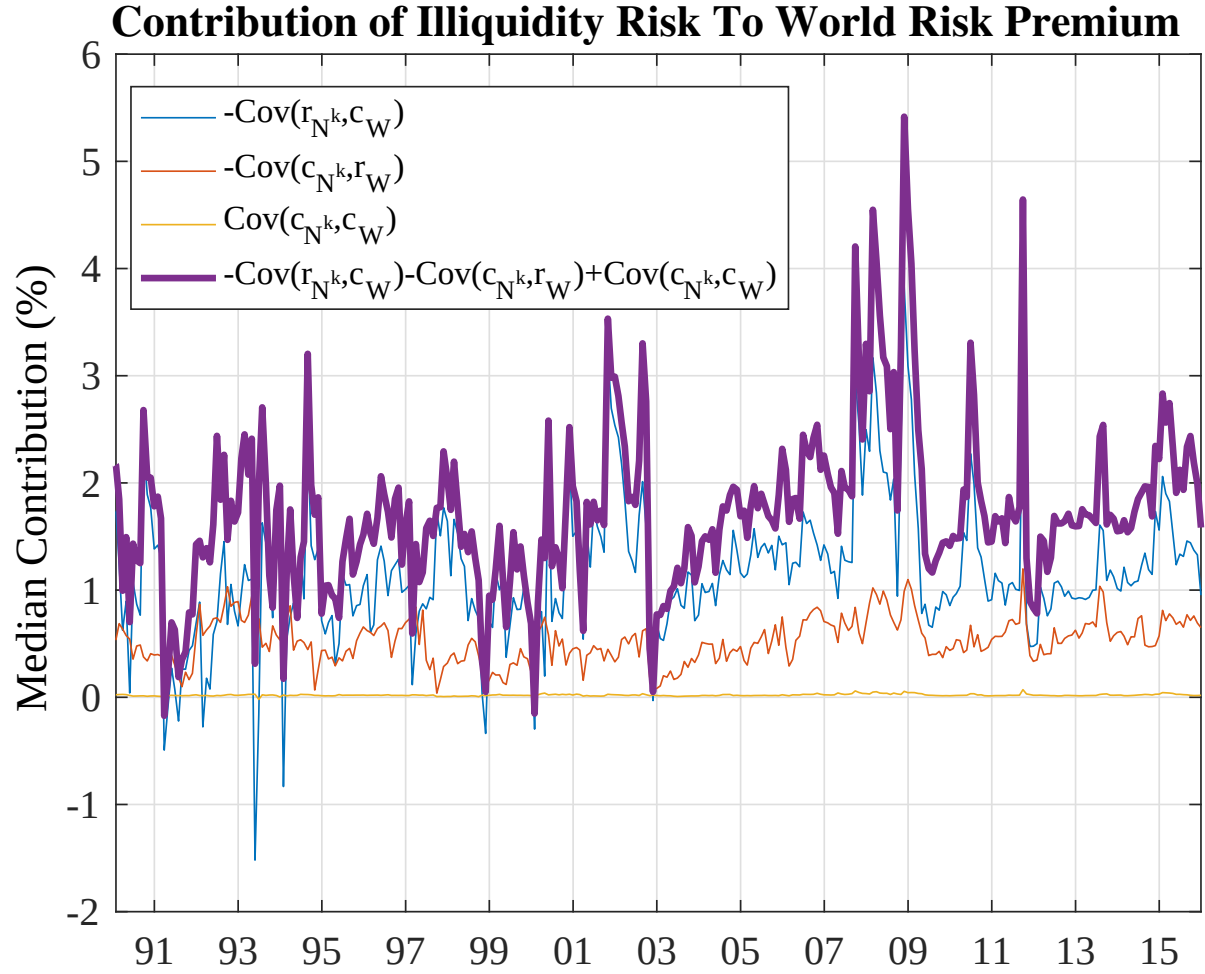


Figure 4 Relative Contribution of Illiquidity Risk to World Risk Premium

We report the median proportion in percentage of each illiquidity risk to total world covariance. The thick line reports the sum of the three illiquidity covariances. We compute the median across all countries available at each point in time.

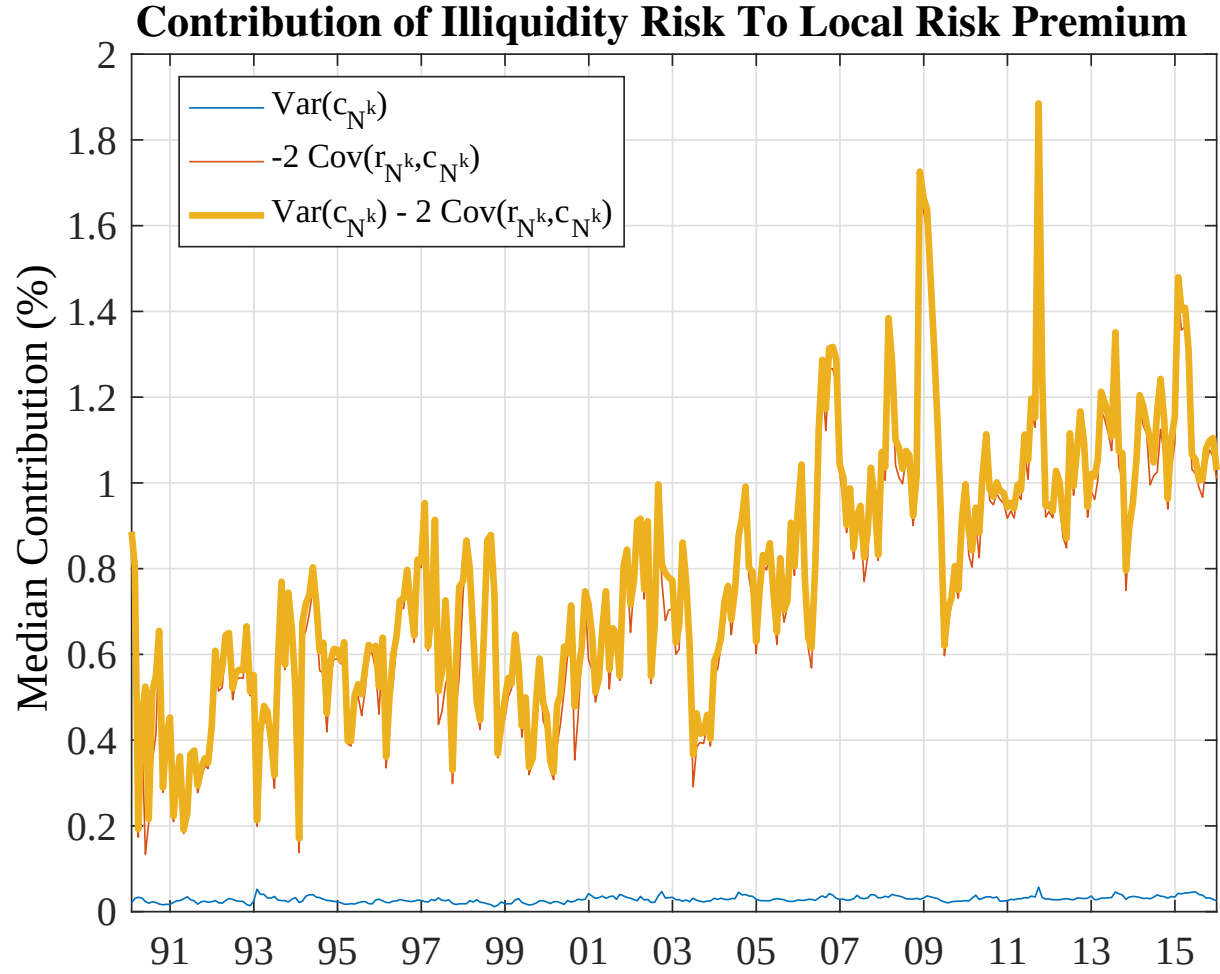


Figure 5 Relative Contribution of Illiquidity Risk to Local Risk Premium

We report the median proportion in percentage of the each illiquidity risk to total local variance. The thick line reports the sum of the two illiquidity risks. We compute the median across all countries available at each point in time.

Table 1 - The number of stocks and the summary statistics of returns and liquidity measure by country and year.

For each country, we report the total number of local stocks over the whole period (column (1)), the number of local stocks at the start of the sample and every five years (columns (2)-(9)), the average and volatility of the returns and bid-ask spread (columns (10)-(13)). We report each statistic for the portfolio of local stocks, of local non-investable stocks, and local investable stocks. For each country, we report in italics in column (1) the proportion of investable stocks and in column (11) the ratio of the average bid-ask spreads of the non-investable portfolio to the investable portfolio. The average returns and return volatilities are annualized. All portfolios are constructed from individual stocks as detailed in the Data section of the paper. Returns are denominated in USD and in excess of the one-month US T-bill rate. The bid-ask spread is the Corwin and Schultz (2012) approximation using high and low prices. The averages and volatility are computed over the sample period when both the non-investable and the investable portfolios are available.

Country	Number of stocks in the sample									Average (%)		Volatility (%)	
	N	1988	1993	1998	2003	2008	2013	2014	2015	Return	Bid-ask spread	Return	Bid-ask spread
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Argentina	<i>17.58%</i>										<i>1.15</i>		
All	91	0	38	26	37	28	39	31	36	3.58	0.60	37.64	0.26
Non-Investable	88	0	36	21	31	19	28	20	25	1.86	0.65	40.53	0.24
Investable	16	0	2	5	6	9	11	11	11	4.99	0.57	39.03	0.30
Brazil	<i>7.64%</i>										<i>1.22</i>		
All	275	0	0	22	42	118	158	152	147	8.04	0.85	38.23	0.20
Non-Investable	267	0	0	21	33	105	141	135	130	8.55	0.94	39.98	0.21
Investable	21	0	0	1	9	13	17	17	17	6.56	0.77	43.01	0.29
Chile	<i>10.22%</i>										<i>0.70</i>		
All	137	0	2	23	33	40	45	43	35	2.84	0.47	23.46	0.16
Non-Investable	129	0	2	17	24	31	38	37	29	3.52	0.41	24.89	0.14
Investable	14	0	0	6	9	9	7	6	6	2.46	0.59	24.01	0.28
China	<i>7.72%</i>										<i>1.04</i>		
All	3017	0	96	786	1260	1562	2336	2162	2219	4.82	0.91	35.02	0.26
Non-Investable	2832	0	86	708	1117	1377	2140	1974	2036	5.52	0.92	36.35	0.27
Investable	233	0	10	78	143	185	196	188	183	3.81	0.89	35.06	0.30
Egypt	<i>6.17%</i>										<i>1.20</i>		
All	162	0	0	44	45	87	98	96	74	0.94	0.78	29.62	0.24
Non-Investable	156	0	0	41	40	82	95	93	70	- 1.72	0.82	29.03	0.27
Investable	10	0	0	3	5	5	3	3	4	6.28	0.69	36.93	0.27
Greece	<i>8.61%</i>										<i>1.25</i>		
All	418	16	84	183	302	204	101	82	62	- 8.13	0.97	39.40	0.33

Country	Number of stocks in the sample									Average (%)		Volatility (%)	
	N	1988	1993	1998	2003	2008	2013	2014	2015	Return	Bid-ask spread	Return	Bid-ask spread
Non-Investable	415	16	84	170	282	181	81	63	43	- 4.44	1.10	38.65	0.26
Investable	36	0	0	13	20	23	20	19	19	- 8.42	0.88	40.85	0.38
Hungary	42.42%										1.20		
All	66	0	3	22	27	22	22	19	19	4.67	0.67	33.44	0.22
Non-Investable	52	0	3	7	9	11	9	8	10	0.60	0.80	36.74	0.27
Investable	28	0	0	15	18	11	13	11	9	4.86	0.66	33.70	0.22
India	0.92%										1.04		
All	2937	0	183	149	538	810	510	952	776	2.68	1.22	37.56	0.25
Non-Investable	2937	0	183	147	537	804	499	948	775	2.07	1.22	37.94	0.25
Investable	27	0	0	2	1	6	11	4	1	- 2.92	1.18	62.88	0.48
Indonesia	15.97%										1.29		
All	501	0	39	85	95	91	234	256	215	- 0.61	1.21	44.67	0.84
Non-Investable	481	0	39	83	73	41	174	190	158	- 3.17	1.38	45.68	1.01
Investable	80	0	0	2	22	50	60	66	57	1.36	1.07	45.18	0.72
Israel	7.72%										1.59		
All	557	0	27	87	104	154	198	207	177	6.75	0.51	23.42	0.18
Non-Investable	532	0	26	83	91	135	178	186	156	4.34	0.60	27.05	0.23
Investable	43	0	1	4	13	19	20	21	21	20.59	0.38	27.09	0.15
Malaysia	1.20%										0.95		
All	998	102	295	379	575	271	458	500	464	3.16	0.66	31.09	0.23
Non-Investable	998	96	286	372	571	270	457	499	463	3.09	0.66	31.41	0.23
Investable	12	6	9	7	4	1	1	1	1	3.71	0.70	31.58	0.27
Mexico	26.36%										1.04		
All	220	13	53	46	41	49	65	68	77	8.47	0.59	30.95	0.22
Non-Investable	200	12	50	37	29	10	27	29	38	9.93	0.59	31.91	0.21
Investable	58	1	3	9	12	39	38	39	39	4.37	0.57	32.28	0.24
Morocco	4.76%										1.44		
All	84	0	0	17	11	42	15	19	14	4.02	0.48	15.70	0.11
Non-Investable	83	0	0	16	9	39	13	17	13	4.60	0.58	16.80	0.14
Investable	4	0	0	1	2	3	2	2	1	3.38	0.40	17.11	0.13
Philippines	12.24%										1.63		
All	245	1	43	46	24	52	91	131	118	4.21	0.68	30.36	0.16

Country	Number of stocks in the sample									Average (%)		Volatility (%)	
	N	1988	1993	1998	2003	2008	2013	2014	2015	Return	Bid-ask spread	Return	Bid-ask Spread
Non-Investable	218	1	33	41	20	48	89	127	117	2.78	0.76	31.64	0.20
Investable	30	0	10	5	4	4	2	4	1	6.79	0.46	31.63	0.25
Poland	3.30%										1.08		
All	849	0	1	2	62	277	481	455	455	7.83	0.77	31.01	0.19
Non-Investable	841	0	1	1	61	268	464	435	432	8.14	0.80	30.99	0.18
Investable	28	0	0	1	1	9	17	20	23	8.52	0.74	32.35	0.25
Russia	20.82%										1.88		
All	293	0	0	7	19	99	117	106	122	5.85	0.93	49.05	0.53
Non-Investable	262	0	0	3	5	60	82	73	90	13.77	1.65	48.11	2.85
Investable	61	0	0	4	14	39	35	33	32	4.30	0.88	49.85	0.46
South Africa	15.18%										0.98		
All	731	0	39	254	142	162	183	191	181	3.68	0.70	27.23	0.28
Non-Investable	684	0	14	226	117	107	131	142	134	3.16	0.68	27.73	0.25
Investable	111	0	25	28	25	55	52	49	47	5.03	0.70	28.08	0.29
South Korea	1.03%										1.40		
All	2803	402	642	709	1480	1699	1673	1714	1800	- 0.95	0.98	37.47	0.30
Non-Investable	2800	402	642	703	1466	1681	1659	1700	1786	- 3.20	1.10	40.36	0.32
Investable	29	0	0	6	14	18	14	14	14	4.19	0.78	35.55	0.28
Taiwan	5.83%										1.13		
All	1029	0	236	444	668	704	819	825	824	- 0.19	0.70	27.06	0.18
Non-Investable	1029	0	236	432	628	681	786	794	793	- 0.42	0.73	27.26	0.18
Investable	60	0	0	12	40	23	33	31	31	0.22	0.64	30.04	0.19
Thailand	7.23%										1.11		
All	802	66	259	110	250	266	429	486	504	2.49	0.81	36.99	0.26
Non-Investable	787	66	259	86	216	225	385	442	461	1.76	0.87	37.15	0.27
Investable	58	0	0	24	34	41	44	44	43	2.92	0.78	37.74	0.26
Turkey	8.50%										1.07		
All	494	42	136	261	276	298	373	368	367	4.73	1.01	44.86	0.37
Non-Investable	493	42	136	253	246	275	366	362	364	6.24	1.03	44.08	0.35
Investable	42	0	0	8	30	23	7	6	3	2.86	0.97	46.70	0.40
All Countries		642	2176	3702	6031	7035	8445	8863	8686				
Proportion of investable		1.1%	2.8%	6.3%	7.1%	8.3%	7.1%	6.6%	6.5%				

Table 2: Pricing tests of the model

The estimated model is:

$$\begin{aligned}
r_{W,t} - c_{W,t} - r_{f,t} + \frac{1}{2}\sigma_{W,t}^2 &= \alpha_W + \kappa_W E[c_{W,t}] + \gamma_{t-1}\sigma_{W,t}^2 + \epsilon_{W,t} \\
r_{N^k,t} - c_{N^k,t} - r_{f,t} + \frac{1}{2}\sigma_{N^k,t}^2 &= \alpha_{N^k} + \kappa_{N^k} E[c_{N^k,t}] + \gamma_{t-1}\Sigma_{N^k,W,t} \\
&\quad + \pi_{t-1}^k \sigma_{N^k,t}^2 \left(1 - \frac{\Sigma_{N^k,DP^k,t}^2}{\sigma_{N^k,t}^2 \sigma_{DP^k,t}^2}\right) + \epsilon_{N^k,t} \\
r_{DP^k,t} - c_{DP^k,t} - r_{f,t} + \frac{1}{2}\sigma_{DP^k,t}^2 &= \alpha_{DP^k} + \kappa_{DP^k} E[c_{DP^k,t}] + \gamma_{t-1}\Sigma_{DP^k,W,t} + \epsilon_{DP^k,t}
\end{aligned}$$

where $r_{N^k,t}$ is the return on the portfolio of non-investables of country k , $r_{DP^k,t}$ is the corresponding diversification portfolio return, $r_{W,t}$ is the return on the global market portfolio, γ_W is the price of world covariance risk, π^k are the prices of conditional local risk, α are intercepts and $\epsilon_t | \mathcal{G}_{t-1} \sim N(0, H_t)$. The time-varying prices are estimated with a different set of conditioning information. Price specifications are given by, $\gamma_t = (\lambda_W' \mathbf{Z}_{W,t-1})^2$, where $\mathbf{Z}_W$ is a set of information variables which includes a constant, the U.S. default spread, the U.S. term structure spread and the world dividend yield in excess of the risk free rate, and $\pi_t^k = (\lambda_k' \mathbf{Z}_{k,t-1})^2$ where $\mathbf{Z}_k$ is a set of local information variables for country k which includes a constant, the lagged local equity return, the local dividend yield and the change in the local exchange rate.

Σ_t is the time-varying conditional covariance parameterized as, $\Sigma_t = \Sigma_0 \circ (\mathbf{1}\mathbf{1}' - \mathbf{a}\mathbf{a}' - \mathbf{b}\mathbf{b}') + \mathbf{b}\mathbf{b}' \circ \Sigma_{t-1} + \mathbf{a}\mathbf{a}' \circ \epsilon_{t-1}\epsilon_{t-1}'$,

where $\circ$ denotes the Hadamard product, $\mathbf{a}$ and $\mathbf{b}$ are (3×1) vector of constants, $\mathbf{1}$ is (3×1) unit vector, and Σ_{t-1} is the matrix of cross error terms, $\epsilon_{t-1}\epsilon_{t-1}'$.

The portfolio of non-investables is constructed from stock level data where a security is non-investable if no ADR or direct listing is available for that security, and the world market portfolio is a value-weighted portfolio of all stocks. The risk free rate is the one-month T-bill rate from French website. All returns are denominated in USD. Sample is from January 1989 or later to December 2015 except for Greece were the sample stops on October 2008. The model is estimated by Quasi-Maximum Likelihood. Panel A reports the specification tests of the world covariance risk. Column (1) reports the starting date of the estimation of world market equation. Columns (2)-(7) report Wald tests and below their p-values for the test hypotheses of the three model specifications; (1) net returns and $\kappa = 0$, (2) net returns and unconstrained κ , (3) gross returns obtained by setting $c = 0$. Columns (8)-(10) report the intercepts of world market equation from the three model specifications.

Panel B reports the conditional local price of risk tests from the main and benchmark specifications; net returns & $\kappa = 0$, and gross returns. Column (1) reports the starting date of the model estimation for each country. The next four columns (2)-(5) report Wald tests and below their p-values for the test hypotheses. Columns

Panel A. Specification tests of world market covariance risk

Null hypothesis	Start date of estimation	for insignificant world market risk ($\lambda_{i,W}=0$ for all i) (d.f. = 4)			for constant world market risk ($\lambda_{i,W}=0$ for $i>1$) (d.f.=3)			Intercept		
		net returns & $\kappa = 0$	net returns & unconstrained κ	gross returns	net returns & $\kappa = 0$	net returns & unconstrained κ	gross returns	net returns & $\kappa = 0$	unconstrained κ	gross returns
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
World market	April-30-1984	2957.35	6343.36	3536.30	394.44	496.51	429.48	0.00	0.03	0.00
		0.00	0.00	0.00	0.00	0.00	0.00	0.44	2.40	0.86

Panel B. Specification tests of conditional local market risk

Null hypothesis	Start date of estimation	for insignificant local market risk ($\lambda_{i,k}=0$ for all i) (d.f. = 4)		for constant local market risk ($\lambda_{i,k}=0$ for $i>1$) (d.f.=3)		Intercept			
		net returns & $\kappa = 0$	gross returns	net returns & $\kappa = 0$	gross returns	net returns & $\kappa = 0$		gross returns	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Argentina	November-30-1993	11943.69	11166.03	6270.66	2240.15	0.00	0.01	-0.04	0.00
		0.00	0.00	0.00	0.00	0.57	1.36	1.76	0.62
Brazil	November-30-1994	5647.56	5645.57	3643.06	4601.33	0.00	0.00	0.01	0.00
		0.00	0.00	0.00	0.00	0.22	0.44	1.76	0.12
Chile	July-31-1995	25866.08	37885.79	737.19	3171.19	-0.02	0.00	-0.01	0.00
		0.00	0.00	0.00	0.00	2.11	0.54	1.52	0.73
China	September-30-1993	3157.10	3410.32	3126.61	3263.87	-0.01	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	1.31	0.40	0.26	0.54
Egypt	January-31-2001	7152.52	4367.10	6511.21	2406.15	-0.01	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	1.44	0.65	0.14	0.29
Greece	March-30-1990	109.01	128.78	14.63	16.03	0.00	0.00	0.01	0.00
		0.00	0.00	0.00	0.00	0.06	0.82	1.52	1.01
Hungary	June-30-1995	1125.81	1402.11	267.97	411.17	-0.01	0.00	-0.01	0.00
		0.00	0.00	0.00	0.00	1.07	0.31	0.67	0.15
India	October-30-1992	6392.64	7591.82	1258.61	1541.76	-0.05	0.00	-0.05	0.00
		0.00	0.00	0.00	0.00	1.95	0.82	1.73	0.96
Indonesia	November-30-1990	71148.30	91447.67	25926.16	20941.86	-0.01	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.99	0.19	0.48	0.40
Israel	April-30-1993	2659.76	3407.95	568.18	879.68	-0.02	0.00	-0.02	0.00
		0.00	0.00	0.00	0.00	1.41	0.20	1.20	0.66

Malaysia	May-30-1986	31586.18	37651.59	31569.23	6733.30	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.11	0.05	1.70	0.05
Mexico	July-31-1989	225150.53	3003.06	20031.82	340.97	0.00	0.01	0.00	0.01
		0.00	0.00	0.00	0.00	1.32	2.43	0.05	2.25
Morocco	January-31-2001	3546.29	4759.01	526.86	679.30	-0.05	0.00	-0.06	0.00
		0.00	0.00	0.00	0.00	1.60	0.12	1.70	0.07
Philippines	April-30-1990	19271.37	17620.30	4662.17	4122.82	-0.01	0.00	-0.01	0.00
		0.00	0.00	0.00	0.00	1.96	0.56	0.99	0.22
Poland	March-30-2001	7206.87	7510.90	4898.28	3389.99	-0.01	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	1.29	0.60	0.09	1.60
Russia	January-31-2001	8585.01	9664.65	2761.80	3308.17	0.00	0.00	0.01	0.00
		0.00	0.00	0.00	0.00	0.22	0.41	1.03	0.11
South Africa	April-30-1990	3140.68	7346.49	439.29	1105.26	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.30	0.39	0.51	0.80
South Korea	November-30-1987	183.00	430.08	26.58	68.28	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.97	0.00	0.85	0.26
Taiwan	October-31-1989	1674.90	2307.93	359.84	451.30	-0.01	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	1.83	1.32	0.05	1.17
Thailand	May-29-1987	9202.45	12419.12	5927.82	9080.03	0.00	0.00	0.01	0.00
		0.00	0.00	0.00	0.00	0.10	0.48	1.30	0.14
Turkey	August-31-1989	14609.82	17358.01	7279.05	7399.56	-0.01	0.00	0.00	-0.01
		0.00	0.00	0.00	0.00	1.15	0.85	0.03	1.15

Table 3: Robustness tests of the model

The estimated model is:

$$\begin{aligned}
 r_{W,t} - c_{W,t} - r_{f,t} + \frac{1}{2}\sigma_{W,t}^2 &= \alpha_W + \kappa_W E[c_{W,t}] + \gamma_{t-1}\sigma_{W,t}^2 + \epsilon_{W,t} \\
 r_{N^k,t} - c_{N^k,t} - r_{f,t} + \frac{1}{2}\sigma_{N^k,t}^2 &= \alpha_{N^k} + \kappa_{N^k} E[c_{N^k,t}] + \gamma_{t-1}\Sigma_{N^k,W,t} \\
 &\quad + \pi_{t-1}^k \sigma_{N^k,t}^2 \left(1 - \frac{\Sigma_{N^k,DP^k,t}^2}{\sigma_{N^k,t}^2 \sigma_{DP^k,t}^2} \right) + \epsilon_{N^k,t} \\
 r_{DP^k,t} - c_{DP^k,t} - r_{f,t} + \frac{1}{2}\sigma_{DP^k,t}^2 &= \alpha_{DP^k} + \kappa_{DP^k} E[c_{DP^k,t}] + \gamma_{t-1}\Sigma_{DP^k,W,t} + \epsilon_{DP^k,t}
 \end{aligned}$$

where $r_{MN^k,t}$ is the return on the portfolio of non-investables of country k , $r_{DP^k,t}$ is the corresponding diversification portfolio return, $r_{W,t}$ is the return on the global market portfolio, γ_W is the price of world covariance risk, π^k are the prices of conditional local risk, α are intercepts and $\epsilon_t | \mathcal{H}_{t-1} \sim N(0, H_t)$. The time-varying prices are estimated with a different set of conditioning information. Price specifications are given by, $\gamma_t = (\lambda_W' \mathbf{Z}_{W,t-1})^2$, where $\mathbf{Z}_W$ is a set of information variables which includes a constant, the U.S. default spread, the U.S. term structure spread and the world dividend yield in excess of the risk free rate, and $\pi_t^k = (\lambda_k' \mathbf{Z}_{k,t-1})^2$ where $\mathbf{Z}_k$ is a set of local information variables for country k which includes a constant, the lagged local equity return, the local dividend yield and the change in the local exchange rate.

Σ_t is the time-varying conditional covariance parameterized as, $\Sigma_t = \Sigma_0 \circ (\mathbf{1}\mathbf{1}' - \mathbf{a}\mathbf{a}' - \mathbf{b}\mathbf{b}') + \mathbf{b}\mathbf{b}' \circ \Sigma_{t-1} + \mathbf{a}\mathbf{a}' \circ \epsilon_{t-1}\epsilon_{t-1}'$,

where $\circ$ denotes the Hadamard product, $\mathbf{a}$ and $\mathbf{b}$ are (3×1) vector of constants, $\mathbf{1}$ is (3×1) unit vector, and Σ_{t-1} is the matrix of cross error terms, $\epsilon_{t-1}\epsilon_{t-1}'$. The portfolio of non-investables is constructed from stock level data where a security is non-investable if no ADR or direct listing is available for that security, and the world market portfolio is a value-weighted portfolio of all stocks. The risk free rate is the one-month T-bill rate from French website. All returns are denominated in USD. Sample is from January 1989 or later to December 2015 except for Greece where the sample stops on October 2008. The model is estimated by Quasi-Maximum Likelihood.

Panel A reports the specification tests when leaving unconstrained κ . Column (1) reports the starting date of the estimation for each country k . Columns (2) and (3) report Wald tests and below their p-values for the test hypotheses of significant and time-varying price of conditional local market risk. Columns (4) and (5) report the intercepts of non-investables and their DPs from the two model specifications. Column (6) reports a Likelihood Ratio (LR) test computed as the difference between the likelihoods of the unconstrained κ model and the zero κ model. Columns (7) and (8) report the estimated free parameter κ . Significant values at the 5% level are bolded.

Panel B reports the specification tests from the main and benchmark specifications; net returns & $\kappa = 0$, and gross returns when assuming all securities are non-investables. Column (1) reports the starting date of the estimation for each country k . The next four columns (2)-(5) report Wald tests and below their p-values for the test hypotheses of significant and time-varying price of conditional local market risk. Columns (6)-(9) report the intercepts of non-investables and their DPs from the two model specifications. Significant values at the 5% level are bolded.

Panel A. Alternative model specification with unconstrained κ

Null hypothesis	Start date of estimation	for insignificant local market risk ($\lambda_{i,k}=0$ for all i) (d.f. = 4)	for constant local market risk ($\lambda_{i,k}=0$ for $i>1$) (d.f.=3)	Intercept		LR test	Free parameter κ	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Argentina	November-30-1993	11926.54 0.00	8330.57 0.00	0.03 1.18	0.00 0.05	11.61	-6.79 1.82	-0.26 0.16
Brazil	November-30-1994	4804.67 0.00	4789.92 0.00	0.04 2.09	0.01 0.80	16.67	-5.84 2.62	-2.97 2.93
Chile	July-31-1995	5769.18 0.00	1544.36 0.00	0.03 2.41	0.00 0.22	10.40	-8.67 3.25	-0.57 0.60
China	September-30-1993	3339.10 0.00	3338.12 0.00	-0.04 1.62	0.00 0.11	5.23	3.35 1.08	-1.00 0.95
Egypt	January-31-2001	5457.76 0.00	394.71 0.00	0.01 0.19	0.06 4.00	2.87	-6.31 2.29	-10.27 4.47
Greece	March-30-1990	231.45 0.00	26.99 0.00	0.01 0.55	0.01 3.86	-2.01	-1.74 0.93	-3.10 6.36
Hungary	June-30-1995	1286.20 0.00	302.98 0.00	0.01 0.50	0.05 3.63	0.16	-4.60 1.48	-8.47 4.66
India	October-30-1992	5592.94 0.00	1294.92 0.00	-0.04 1.33	0.00 0.23	3.82	-2.10 1.04	-1.98 1.96
Indonesia	November-30-1990	70480.96 0.00	23678.67 0.00	0.01 0.61	0.00 0.61	4.91	-1.87 2.13	-1.55 2.18
Israel	April-30-1993	3504.74 0.00	399.79 0.00	-0.02 1.21	0.00 1.19	-1.66	-2.51 1.56	-2.37 3.28
Malaysia	May-30-1986	32105.25 0.00	5113.28 0.00	0.01 0.99	0.00 0.65	-1.00	-2.28 1.89	-0.38 0.69

Mexico	July-31-1989	3976.35	2539.53	0.02	0.03	6.77	-4.82	-7.09
		0.00	0.00	2.22	4.72		3.50	6.97
Morocco	January-31-2001	4817.52	576.64	-0.07	-0.01	2.79	-0.39	3.29
		0.00	0.00	1.83	0.91		0.12	0.96
Philippines	April-30-1990	13813.31	2935.80	0.02	0.00	0.04	-5.18	-0.79
		0.00	0.00	0.62	0.19		1.32	0.65
Poland	March-30-2001	7799.25	5730.07	0.02	0.00	-3.00	-4.45	-1.94
		0.00	0.00	0.82	0.53		1.48	2.08
Russia	January-31-2001	7209.38	1877.15	0.01	0.01	3.58	-2.33	-3.24
		0.00	0.00	0.64	1.61		1.48	2.98
South Africa	April-30-1990	22201.38	4492.79	0.02	-0.01	11.48	-5.32	-0.38
		0.00	0.00	2.19	1.13		4.19	0.52
South Korea	November-30-1987	75.06	11.97	0.00	-0.01	6.17	-0.98	-0.43
		0.00	0.01	0.16	1.33		0.90	0.58
Taiwan	October-31-1989	3483.53	505.35	0.01	0.00	10.55	-3.46	-2.04
		0.00	0.00	0.80	0.25		1.70	1.64
Thailand	May-29-1987	8308.60	5032.57	0.01	0.01	-0.38	-2.29	-1.90
		0.00	0.00	0.85	0.88		1.44	1.96
Turkey	August-31-1989	16407.88	7673.39	0.02	0.01	7.94	-3.74	-2.46
		0.00	0.00	0.94	0.47		2.12	1.74

Panel B. Specification tests when all securities are non-investables

Null hypothesis	Start date of estimation	for insignificant local market risk ($\lambda_{i,k}=0$ for all i) (d.f. = 4)		for constant local market risk ($\lambda_{i,k}=0$ for $i>1$) (d.f.=3)		Intercept			
		net returns & $\kappa = 0$	gross returns	returns & $\kappa = 0$	gross returns	net returns & $\kappa = 0$		gross returns	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Argentina	November-30-1993	1456.66	8994.74	217.69	916.14	0.00	0.00	0.01	0.00
		0.00	0.00	0.00	0.00	0.40	0.71	1.03	0.60
Brazil	November-30-1994	11728.36	9452.46	893.47	1060.03	0.00	0.00	0.01	0.00
		0.00	0.00	0.00	0.00	0.08	0.52	1.84	0.05
Chile	July-31-1995	3610.32	12124.95	233.75	1749.42	-0.01	0.00	-0.02	0.00
		0.00	0.00	0.00	0.00	0.71	0.43	1.49	0.75
China	September-30-1993	1243.34	1370.38	1023.55	955.57	-0.01	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.75	0.32	0.36	0.04
Egypt	January-31-2001	8758.45	8516.92	517.73	543.38	-0.05	0.00	-0.05	0.00
		0.00	0.00	0.00	0.00	1.72	0.60	1.54	0.43
Greece	March-30-1990	1142.99	923.01	450.45	233.24	0.00	0.00	0.01	0.00
		0.00	0.00	0.00	0.00	0.20	1.33	2.10	1.73
Hungary	June-30-1995	7935.07	8489.63	699.91	760.34	-0.01	0.01	0.00	0.00
		0.00	0.00	0.00	0.00	1.27	1.44	0.15	0.98
India	October-30-1992	9783.34	11327.23	1966.28	2307.87	-0.06	0.00	-0.06	0.00
		0.00	0.00	0.00	0.00	2.05	0.09	1.91	0.32
Indonesia	November-30-1990	6524.14	35156.91	4675.02	32493.24	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.27	0.18	1.21	0.54
Israel	April-30-1993	627.58	754.34	294.55	472.80	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.16	0.16	1.46	0.29

Malaysia	May-30-1986	202696.47	153754.47	92022.17	77265.10	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.72	0.24	1.28	0.34
Mexico	July-31-1989	273778.90	196501.58	36059.70	40336.67	0.01	0.01	0.01	0.01
		0.00	0.00	0.00	0.00	1.90	2.24	3.21	1.62
Morocco	January-31-2001	2641.73	2801.46	1898.16	2043.11	-0.01	0.00	-0.01	0.00
		0.00	0.00	0.00	0.00	1.55	0.88	0.88	1.47
Philippines	April-30-1990	16597.77	15634.77	2752.81	2473.34	-0.01	0.00	-0.01	0.00
		0.00	0.00	0.00	0.00	1.77	0.09	0.90	0.42
Poland	March-30-2001	4016.51	4526.14	3704.64	4199.61	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.80	0.88	0.63	0.47
Russia	January-31-2001	12735.22	18003.92	2803.58	3013.35	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.61	0.30	0.51	0.26
South Africa	April-30-1990	3557.10	4641.49	285.42	674.74	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	1.14	0.21	0.35	0.25
South Korea	November-30-1987	5664.05	7186.03	741.64	809.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	1.39	0.10	0.84	0.38
Taiwan	October-31-1989	2872.62	3039.03	1922.05	976.51	-0.01	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	2.20	0.06	0.81	0.36
Thailand	May-29-1987	5209.35	6815.01	817.41	626.61	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.21	0.79	1.28	0.46
Turkey	August-31-1989	14986.06	17901.57	1486.73	1775.53	-0.01	0.00	-0.01	0.00
		0.00	0.00	0.00	0.00	1.78	0.30	0.98	0.77